

HANK Made Easy

—Macro Fluctuations and Policies in THANK—

(Tractable Heterogeneous-Agent New Keynesian Models)

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Lecture Notes on THANK, this version June 2021¹

¹Notes used to teach advanced mini-courses over the last years at European Central Bank Research Department, Paris School of Economics Summer School and APE program, HEC Lausanne, etc.

Inspiring Quotes

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3. *"The art of doing mathematics is finding that special case that contains all the germs of generality."* **David Hilbert**
4. *"If you can't solve a problem, then there is an easier problem you can solve: find it."* **George Polya**

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1. *"Like all people who tried to exhaust a subject, he exhausted his listeners"* **Oscar Wilde**
2. *"If I have not seen as far as others, it is because there were giants standing on my shoulders."* **Hal Abelson**

Disclaimers

1. Heterogeneity in ... many things, here **only households** and **very limited**
 - ▶ Precisely why **tractable**: imagine non-tractable even with such a limited scope
 2. **NOT** an exhaustive review of HANK
 - ▶ Centered on *my own work* within this paradigm (15+ yrs & what I *happen to* know best—no doubt many know it better)
 - ▶ + related contributions by others
- ▶ Overlap with *the slides I use to present my research*

Motivation

- ▶ **2008 Great Expansion**—*stabilization* policies (mon&fisc)
 - ▶ + inequality-redistribution, i.a. Bernanke, Yellen, Draghi
- ▶ Micro data & solving HA models Krusell Smith, Den Haan, Reiter (...)
- ▶ *Aggregate* Euler? Hall; Cambell Mankiw (...) **zero net worth:** Wolff (...)
- ▶ Consumption—Income: Johnson, Parker, Souleles; Surico et al; etc.
- ▶ *Liquidity* constr. & MPC: Kaplan Violante; Cloyne Ferreira Surico; Gorea Midrigan

HA

- ▶ Heterogeneity and constraints: many (parallel and often deeply similar) ways throughout the decades
 - ▶ HANK is the culmination–synthesis
- ▶ Ex: Bewley-Aiyagari-Huggett (Imrohoroglu, Krusell Smith, Rios-Rull, Heathcote, etc.)
- ▶ vs. Deaton-Carroll-Zeldes (Kimball, Mankiw, Campbell, etc.)
- ▶ Looks like a *divide* and we did not even mention prices yet ... (up to the historians of thought).

$$\frac{HA}{NK}$$

HA
— — — —
HANK

2000s: TANK, Macro to Micro

2010s: Micro to Macro

HA_{NK}

$\overline{HA} \overline{NK}$

2000s: TANK, Macro to Micro

Based on (15+ years)

- ▶ Limited Asset Market Participation, Monetary Policy, and (Inverted) Aggregate Demand Logic, **2008** *Journal of Economic Theory* (Ch. 1, **2004** PhD Thesis)
- ▶ The New Keynesian Cross, 2017a *Journal of Monetary Economics*
- ▶ Monetary Policy and Heterogeneity: An Analytical Framework, 2017b *Mimeo*

Please cite & acknowledge the above 3 if using these slides

- ▶ Joint work w/ **R. Straub** (2004 Mimeo, 2012 JEDC, 2013 REStat); **Meier and Mueller** (2008 JMCB); **Monacelli and Perotti** (2011 Mimeo; 2013 EJ); **Ragot** (2016 Mimeo)
- ▶ Ongoing work with **Primiceri and Tambalotti, Känzig and Surico, Monacelli and Perotti, etc.**

Quantitative | Tractable

TANK
2000s

Bilbiie **2008** *JET* (2004)
LAMP, Mon. Pol. & (Inv.) Aggregate Demand

HANK
2010s

χ

Key channel: $\chi \geq 1 \sim$ Cyclical Inequality

Elasticity of individual to aggregate income: $F(\text{profits}' \text{ (re)distribution})$ [link](#)

Some ECB history (disclaimer applies ...)



WORKING PAPER SERIES NO. 408 / NOVEMBER 2004

THE GREAT INFLATION, LIMITED ASSET MARKETS PARTICIPATION AND AGGREGATE DEMAND:

FED POLICY WAS BETTER THAN YOU THINK¹

by Florin O. Bilbie²



In 2004 all
publications
in the ECB
library are
available
electronically
from the
ECB website.

This paper can be downloaded without charge from
<http://www.ecb.int> or from the Social Science Research Network
electronic library at http://ssrn.com/abstract_id=601028.

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WORKING PAPER SERIES NO 1438 / MAY 2012

ASSET MARKET PARTICIPATION, MONETARY POLICY RULES AND THE GREAT INFLATION

by Florin O. Bilbie
and Roland Straub



In 2012 all ECB
publications
published in a paper
format from
the ECB website.

NOTE: This Working Paper should not be reported as representing the
views of the European Central Bank (ECB). The views expressed are
those of the authors and do not necessarily reflect those of the ECB.

Literature

- ▶ **TANK 2000s** Bilbiie 08; Galí Lopez-Salido Vallés 07 (Mankiw 00); Bilbiie Straub; Bilbiie Meier Muller; Colciago; Di Bartolomeo Rossi; Areosa Areosa; Furlanetto; Natvik; Ascari, Colciago and Rossi; Eser, etc.; **different:** Iacoviello 05; Eggertsson Krugman; Curdia Woodford; Nistico; Bilbiie Monacelli Perotti
- ▶ **HANK 2010s** Oh Reis, Guerrieri Lorenzoni, Gornemann Kuester Nakajima; **Kaplan Moll Violante; McKay Nakamura Steinsson; Auclert;** Auclert Rognlie; Bayer Luetticke Pham-Dao Tjaden; Luetticke; Ravn Sterk; Den Haan Rendahl Riegler; McKay Reis; Challe Matheron Ragot Rubio; Debortoli Galí; Hagedorn Manovskii Mitman (Luo); Ferrière Navarro; Auclert Rognlie Straub; Wolf; Fernandez-Villaverde Nuno Rachedi; **Analytical:** Acharya Dogra; Bilbiie; Broer, Hansen, Krusell, Oberg; Holm; Ravn Sterk; Werning; Cantore Freund; Holm; Bernstein; Caramp Silva
- ▶ **Determinacy in RANK:** Leeper; Woodford; Cochrane; Lubik Schorfheide; **Forward Guidance puzzle** (Del Negro, Giannoni, Patterson): ~~perfect information/rational expectations~~ Kiley; Carlstrom Fuerst Paustian; Garcia-Schmidt Woodford; Farhi Werning; Wiederholt; Andrade et al; Gabaix; Angeletos Lian; **G balance sheet:** Cochrane; Diba Loisel; Michaillat Saez; Hagedorn
- ▶ **Optimal policy TANKs** Bilbiie 08, Ascari et al; Nistico; Areosa Areosa; Bilbiie Monacelli Perotti; **HANKs:** Bhandari Evans Golosov Sargent; Nuno Thomas; Challe; Bilbiie Ragot; Cui Sterk; LeGrand Martin-Baillon Ragot;

Core Model: **THANK**

Max(Micro in Macro)

s.t. Tractable

Core Model: **THANK**

1. Projection of **several** quantitative-HANK channels
2. **T**ractable affords *closed-form analytical, full-blown NK*
 - policymakers, central banks
 - public communication
 - students
 - colleague economists
 - discipline empirical work

Plan (all topics about Monetary and Fiscal Policy)

1. What is RANK not enough for?
 - ▶ Aggregate Demand and Keynesian Cross
2. TANK (RANK-isomorphic)
 - ▶ The New Keynesian Cross
3. THANK:
 - ▶ idiosyncratic risk & precautionary saving
 - ▶ liquidity
 - ▶ cyclical risk
 - ▶ (the Catch-22)
4. *Optimal Monetary(-Fiscal) Policies* in TANK & THANK
5. Application: *Liquidity Traps*
Further developments/extensions: THANK &
6. *Money, i.e. Liquidity* (w/ Ragot)
7. Stabilization v Redistribution: *Optimal Monetary-Fiscal Mix* (w/ Monacelli & Perotti)
8. *Capital K, i.e. illiquid wealth* (w/ Känzig & Surico)
9. *DSGE, estimated*, w/ Primiceri & Tambalotti

Preview: The 3-Equation THANK Model

$$c_t = \delta E_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} (i_t - E_t \pi_{t+1} - \rho_t)$$

$$: \quad (\text{with } \delta \equiv 1 + (\chi - 1) \frac{1 - s}{1 - \lambda \chi})$$

$$\pi_t = \kappa c_t + \beta E_t \pi_{t+1} + u_t$$

$$i_t = \phi \pi_t \text{ (or LQ-optimal policy)}$$

► Heterogeneity $\sim$ colors

RANK: A Keynesian-Cross Representation

- Complete markets

$$\max E_0 \sum_{t=0}^{\infty} \beta^t U(C_t^j, N_t^j) \text{ s.t.}$$

$$Z_{t+1}^j + \Theta_{t+1}^j V_t \leq B_t^j + \Theta_t^j (V_t + P_t D_t) + W_t N_t^j - P_t C_t^j.$$

- Z_{t+1}^j nominal *end of period* t portfolio of all state-contingent assets (except shares)
- B_t^j beginning of period wealth. Θ_t^j shares

- **No-arbitrage** $\rightarrow \exists$ pricing kernels/stoch. disc. factors:

$$\frac{Z_{t+1}^j}{P_t} = E_t \left[Q_{t,t+1}^j \frac{B_{t+1}^j}{P_{t+1}} \right] \text{ and } \frac{V_t}{P_t} = E_t \left[Q_{t,t+1}^j \left(\frac{V_{t+1}}{P_{t+1}} + D_{t+1} \right) \right],$$

- Gross **real** rate (definition)

$$\frac{1}{R_t} = E_t Q_{t,t+1}^j$$

RANK: A Keynesian-Cross Representation

- No-arbitrage + wealth $\rightarrow$ flow BC, + 'natural' borrowing limit by state, anticipate equil. all agents hold constant fraction of shares (no trade) Θ^j :

$$\text{IBC:} \quad E_t \sum_{i=0}^{\infty} Q_{t,t+i}^j C_{t+i}^j \leq E_t \sum_{i=0}^{\infty} Q_{t,t+i}^j Y_{t+i}^j$$

$$\text{income} \quad : \quad Y_{t+i}^j \equiv \Theta^j D_{t+i} + \frac{W_{t+i}}{P_{t+i}} N_{t+i}^j$$

Max. U s.t. this, each date and state:

$$\beta \frac{u_C(C_{t+1}^j)}{u_C(C_t^j)} = Q_{t,t+1}^j$$

+ IBC with equality (or flow BC w/ equality + transversality $\lim_{i \rightarrow \infty} E_t [Q_{t,t+i}^j Z_{t+i}^j] = \lim_{i \rightarrow \infty} E_t [Q_{t,t+i}^j V_{t+i}] = 0$).

RANK: A Keynesian-Cross Representation

- Substitute in no-arbitrage

$$\frac{1}{R_t} = \beta E_t \left[\frac{U_C(c_{t+1}^j)}{U_C(c_t^j)} \right].$$

- Loglinearize IBC $E_t \sum_{i=0}^{\infty} Q_{t,t+i}^j c_{t+i}^j \leq E_t \sum_{i=0}^{\infty} Q_{t,t+i}^j Y_{t+i}^j$, use Euler and stochastic disc. factor

$$c_t^j = -\sigma\beta \sum_{i=0}^{\infty} \beta^i E_t r_{t+i} + (1-\beta) \sum_{i=0}^{\infty} \beta^i E_t y_{t+i}^j$$

write in recursive form ...

RANK "not enough" 1: (lack of) amplification

- ▶ "consumption function" of agent j ;

$$c_t^j = (1 - \beta) \hat{y}_t^j - \sigma \beta r_t + \beta E_t c_{t+1}^j.$$

- ▶ Euler-IS by market clearing $c_t^j = \hat{y}_t^j \equiv y_t^j - t_t^j$:

$$c_t^j = E_t c_{t+1}^j - \sigma r_t$$

- ▶ MP shock with persistence p and FP $g_t = t_t$

$$\Omega \equiv \frac{dc_t^j}{d(-r_t)} = \frac{\sigma}{1-p}$$

$$\omega \equiv \frac{\Omega_I}{\Omega} = \frac{1-\beta}{1-\beta p}$$

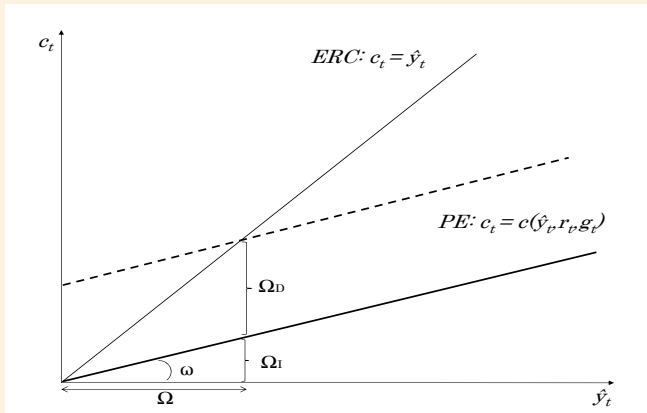
$$\mathcal{M} \equiv \frac{dy_t}{dg_t} = 1 \left(\frac{dc_t}{dg_t} = 0 \right)$$

$$\text{where } \Omega_I \equiv \frac{dc_t^j}{d(-r_t)} \Big|_{r_t=\bar{r}} = \Omega - \Omega_D; \Omega_D \equiv \frac{dc_t^j}{d(-r_t)} \Big|_{y_t^j=\bar{y}} = \frac{\sigma\beta}{1-\beta p}$$

The (New?) Kenesian Cross

$$c_t = \omega \hat{y}_t - (1 - \omega) \Omega r_t + (1 - \omega) (M - 1) g_t$$

Old Keynesian Cross: Samuelson (1948, pp 256-279)



$\omega \sim$ aggreg. MPC $\Omega, M \sim$ Multipliers

RANK: all shift, no slope

- ▶ $\omega \sim 0$ Planned-Expenditure curve is flat
- ▶ – virtually no General-Equilibrium
- ▶ All partial-equilibrium, direct (shift): $\Omega \sim \Omega_D$
- ▶ New ... *what?*

RANK "not enough" 2: FG Puzzle

- add AS $\pi_t = \kappa c_t (+\beta E_t \pi_{t+1})$

$$c_t = E_t c_{t+1} - \sigma (i_t - E_t \pi_{t+1}) = \nu_0 E_t c_{t+1} - \sigma i_t$$

News on AD under i peg:

$$\nu_0 \equiv 1 + \kappa \sigma \geq 1.$$

- **FG puzzle** (Del Negro, Giannoni, Patterson, ...):

$$c_t = \nu_0 E_t c_{t+1} - \sigma i_t = \nu_0^T E_t c_{t+T} - \sigma \sum_{j=0}^{T-1} \nu_0^j E_t i_{t+j}$$

- "indeterminacy", rationale for Taylor $i_t = \phi \pi_t$, $\phi > 1 \longrightarrow$
 $\nu = \frac{1+\kappa\sigma}{1+\kappa\sigma\phi} < 1$, solve $\uparrow$ **forward**

TANK

TANK, Bilbiie 2008 version

- ▶ (Revisited in light of HANK: *The New Keynesian Cross*)
- ▶ Assets or not χ ; λ fraction consume all **their** income

$$C_t^H = W_t N_t^H + \mathcal{T}_t^H = Y_t^H$$

- ▶ S: complete markets

$$C_t^S + Z_{t+1}^S + v_t \Theta_{t+1} = W_t N_t^S + B_t^S + \Theta_t (v_t + D_t) + \mathcal{T}_t^S,$$

- ▶ Assets held/priced, not traded! ($\neq$ Mankiw 2000, Gali et al 2007)
→ isolate role of **income inequality** and profits (#60)

- ▶ Asset mkt clearing→

$$C_t^S = W_t N_t^S + \frac{1}{1-\lambda} D_t + \mathcal{T}_t^S = Y_t^S$$

- ▶ Redistribution/transfer (**Section 4.3**), tax profits τ^D rebate to $H \sim$ "automatic stabilizer"

$$\lambda \mathcal{T}_t^H = \tau^D D_t = -(1-\lambda) \mathcal{T}_t^S$$

- ▶ Heterogeneity in earnings and income

TANK

- ▶ separable $U^j(C^j, N^j)$; $\sigma^{-1} \equiv -U_{CC}^j C^j / U_C^j$; $\varphi \equiv U_{NN}^j N^j / U_N^j$
- ▶ **H work**; S work and trade/price shares (get profits)+all securities. No risk or insurance (later)
- ▶ Labor $\varphi n_t^j = w_t - \sigma^{-1} c_t^j$ all j (also aggregate!) together with $c_t = y_t = n_t \rightarrow$

$$w_t = \left(\varphi + \sigma^{-1} \right) c_t$$

- ▶ H loglin BC $c_t^H = w_t + n_t^H + \frac{\tau^D}{\lambda} d_t - t_t^H$.
- ▶ **government policies**: 1. τ^D ; 2. D^{SS} : steady-state subsidy+tax firms (S) $\rightarrow D^{SS} = 0$

$$d_t = -w_t$$

(now $(g_t, t_t^H) = 0$, add later)

TANK: Model Summary

Equil. condition	Loglinearized
$U_C (C_t^S) = \beta R_t E_t [U_C (C_{t+1}^S)]$	$c_t^S = E_t c_{t+1}^S - \sigma r_t$
$\frac{W_t}{P_t} U_C (C_t^S) = -U_N (N_t^S)$	$\varphi n_t^S = w_t - \sigma^{-1} c_t^S$
$\frac{W_t}{P_t} U_C (C_t^H) = -U_N (N_t^H)$	$\varphi n_t^H = w_t - \sigma^{-1} c_t^H$
$C_t^H = \frac{W_t}{P_t} N_t^H + \frac{\tau^D}{\lambda} D_t$	$c_t^H = w_t + n_t^H + \frac{\tau^D}{\lambda} d_t$
$D_t = (1 + \tau^S) Y_t - \frac{W_t}{P_t} N_t - T_t^F$	$d_t = -w_t$
$Y_t = C_t \equiv \lambda C_t^H + (1 - \lambda) C_t^S$	$y_t = c_t \equiv \lambda c_t^H + (1 - \lambda) c_t^S$
$N_t = \lambda N_t^H + (1 - \lambda) N_t^S$	$n_t = \lambda n_t^H + (1 - \lambda) n_t^S$
$Y_t = N_t$	$y_t = n_t$

► loglin. around SS w/ opt. $\tau^S, D^{SS} = 0, T_t^F = \tau^S Y_t$

TANK: Reminder

- ▶ if prices sticky, add Phillips curve + Taylor rule (π and i)
- ▶ above still goes through with Phillips curve if Taylor rule is
$$i_t = E_t \pi_{t+1} + r_t$$
- ▶ (just as in RANK)

TANK: Key Derivation

- ▶ idea: express individual variables $c^j(=y^j)$ as function of aggregate $c(=y)$; start with:

$$c_t^H = w_t + n_t^H + \frac{\tau^D}{\lambda} d_t$$

replace labor supply $\varphi n_t^H = w_t - \sigma^{-1} c_t^H$, $d_t = -w_t$

$$\left(1 + (\varphi\sigma)^{-1}\right) c_t^H = \left(1 - \frac{\tau^D}{\lambda} + \varphi^{-1}\right) w_t$$

aggregate LS $\varphi n_t = w_t - \sigma^{-1} c_t$ &

$$y_t = n_t = c_t \rightarrow w_t = (\varphi + \sigma^{-1}) y_t$$

- ▶ Define

$$\chi \equiv 1 + \varphi \left(1 - \frac{\tau^D}{\lambda}\right)$$

$$c_t^H = \chi c_t$$

TANK: Deriving Aggregate Demand



$$c_t^H = \underbrace{\chi}_{\text{[KEY]}} y_t = \left[1 + \underbrace{\varphi}_{\text{labor mkt.}} \times \underbrace{\left(1 - \frac{\tau^D}{\lambda}\right)}_{\text{fiscal redistrib.}} \right] y_t$$

$$c_t^S = \frac{1 - \lambda \chi}{1 - \lambda} y_t$$

Extra income effect $w \uparrow \rightarrow d \downarrow$ *keystone*: profits²

- **Gen-Eq.:** demand $\uparrow$, $w\uparrow$, $y^H \uparrow$, demand $\uparrow\uparrow$ -amplification spiral *until?*
- S work more=*equilibrium* because $y^S \downarrow$ as $d \downarrow$

$$\text{Cyclical Income Inequality: } \gamma_t = y_t^S - y_t^H = \frac{1 - \chi}{1 - \lambda} y_t$$

Several income-distribution (χ) models: fiscal incidence, *sticky wages*

TANK: Colciago; Ascari Colciago Rossi, Furlanetto, HANK: Broer et al, Auclert et al.

²Profits' cyclicality? 1. off-equilibrium, 2. conditional; 3. *data!*? 

TANK: Cyclical (Income) Inequality

- ▶ Aggregate Euler-IS-AD: replace c_t^S in Euler S:
 $c_t^S = E_t c_{t+1}^S - \sigma r_t$:

$$c_t = E_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} r_t$$

1. **TANK** Amplification iff $\chi > 1$: **Inequality Countercyclical**

Generalizes to rich-HANK: $\text{cov}(\text{MPC}, \chi)$, Auclert JMP 2015; Direct test: Patterson 2019 JMP

$$\text{agg. MPC} \equiv \lambda \times 1 \times \chi + (1 - \lambda) \times (1 - \beta) \times \frac{1 - \lambda \chi}{1 - \lambda}$$

- ▶ $\chi > 1$: AEIS— dc/dr —*increasing* with λ ($< \chi^{-1}$); Reason $\uparrow$
- ▶ **dampening** with $\chi < 1$ but
 - ▶ indirect share ω increasing with λ *regardless* of χ ;

The New Keynesian Cross

- **Aggreg. C, PE curve** (novel $\neq$ Campbell-Mankiw!):

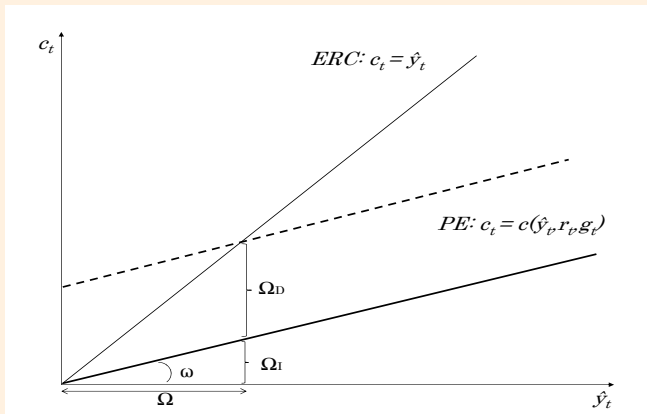
$$c_t = [1 - \beta (1 - \lambda\chi)] \hat{y}_t - (1 - \lambda) \beta \sigma r_t + \beta (1 - \lambda\chi) E_t c_{t+1}$$

- Partial equilibrium, indirect effect ... **MPC!** keep y fixed
- General equilibrium, total effect ... **Multiplier:** add $c_t = \hat{y}_t \rightarrow$ Aggregate Euler

	Total effect Ω ("multiplier")	Indirect-effect share ω ("aggregate MPC")
TANK	$\frac{\sigma}{1-p} \frac{1-\lambda}{1-\lambda\chi}$	$\frac{1-\beta(1-\lambda\chi)}{1-\beta p(1-\lambda\chi)}$

The New Kenesian Cross

$$c_t = \omega \hat{y}_t - (1 - \omega) \Omega r_t + (1 - \omega) (M - 1) g_t$$



$$\text{agg. MPC } \omega \equiv \lambda \times 1 \times \chi + (1 - \lambda) \times (1 - \beta) \times \frac{1 - \lambda \chi}{1 - \lambda}$$

TANK Neutrality Special case: A-cyclical Inequality

- ▶ Campbell-Mankiw knife-edge $\chi = 1$, intertemporal substitution only difference

188 · CAMPBELL & MANKIW

accrues to individuals to consume their current income, while the remainder $(1-\lambda)$ accrues to individuals who consume their permanent income. If the incomes of the two groups are Y_{1t} and Y_{2t} respectively, then total income is $Y_t = Y_{1t} + Y_{2t}$. Since the first group receives λ of total income, $Y_{1t} = \lambda Y_t$ and $Y_{2t} = (1-\lambda)Y_t$. Agents in the first group consume their current income, so $C_{1t} = Y_{1t}$, implying $\Delta C_{1t} = \Delta Y_{1t} = \lambda \Delta Y_t$. By contrast, agents in the second group obey the permanent income hypothesis, implying $\Delta C_{2t} = (1-\lambda)\epsilon_t$.

The change in aggregate consumption can now be written as

$$\Delta C_t = \Delta C_{1t} + \Delta C_{2t} = \lambda \Delta Y_t + (1-\lambda)\epsilon_t. \quad (1.4)$$

- ▶ History of thought: footnote 26 in CM's 3rd and last paper on this, EER 1991

²⁶Of course, utility costs would be much larger again if some agents were consuming their own current income.

- ▶ **neutrality** (RANK); but *indirect effect* (one-to-one);
- ▶ Bilbiie 2008 footnote 14; Bilbiie-Straub 2012;
 - ▶ Werning 2015: generalization in a complicated model but focusing on "income risk". Here, no risk (yet)

Detour: Aggregate Demand, Inverted

- ▶ IS-AD **swivels** when:

$$\lambda > \chi^{-1}$$

- ▶ "fallacy of composition" → Inverted AD & Taylor principle
 - ▶ see also subsequent Di Bartolomeo Rossi on this
- ▶ Bilbiie Straub 2013 REStat explain Great Inflation: no sunspots, passive Fed policy OK (Bayesian TANK-DSGE estimation)
- ▶ Bilbiie Straub 2012 JEDC (1-eq. GMM): IS slope inverted in the 70s, changed sign post-Volcker. Solves zero-slope puzzle (time-aggregate of + and -)
- ▶ Key: tremendous financial liberalization and innovation → increased participation in the early '80s
- ▶ **Takeaway:** to publish "Keynesian" papers with amplification in the 2000s (Great Moderation ...) had to focus on bifurcations, inversions ... non-Keynesian

TANK: Fiscal Multipliers

- ▶ Add **Government policy 3**: spend G_t , balanced-budget $T_t = G_t$
- ▶ + *exogenous* redistribution ($\sim$ progressivity) α

$$\lambda T_t^H = \alpha T_t$$

- ▶ loglinearized (around $G = 0$)

$$t_t^H = \frac{\alpha}{\lambda} t_t = \frac{\alpha}{\lambda} g_t = \underbrace{g_t}_{\text{bal.-budg.}} - \underbrace{\left(1 - \frac{\alpha}{\lambda}\right) t_t}_{\text{exog. redist.}}$$

- ▶ Implies (see derivation next with ζ definition)

$$c_t^H = \chi \hat{y}_t + \zeta \left(\chi - \frac{\alpha}{\lambda} \right) g_t$$

TANK: Fiscal Multipliers Derivation

- Equations that change:

$$y_t = c_t + g_t$$

$$c_t^H = w_t + n_t^H + \frac{\tau^D}{\lambda} d_t - t_t^H$$

replace labor supply $\varphi n_t^H = w_t - \sigma^{-1} c_t^H$, $d_t = -w_t$

$$\left(1 + (\varphi\sigma)^{-1}\right) c_t^H = \left(1 - \frac{\tau^D}{\lambda} + \varphi^{-1}\right) w_t - t_t^H$$

aggregate LS $\varphi n_t = w_t - \sigma^{-1} c_t$ &

$$y_t = n_t = c_t + g_t \rightarrow w_t = (\varphi + \sigma^{-1}) c_t + \varphi g_t$$

- Letting again $\chi = 1 + \varphi \left(1 - \frac{\tau^D}{\lambda}\right)$ and income effect on H

$$\zeta \equiv 1 / \left(1 + (\varphi\sigma)^{-1}\right)$$

$$c_t^H = \chi c_t + \chi \zeta g_t - \zeta t_t^H$$

TANK: Fiscal Multipliers

- Aggregate Euler-IS-AD:

$$c_t = E_t c_{t+1} - \frac{1-\lambda}{1-\lambda\chi} \sigma r_t + \frac{\lambda\zeta}{1-\lambda\chi} \left(\chi - \frac{\alpha}{\lambda} \right) (g_t - E_t g_{t+1}).$$

- *Aggreg. C, PE curve:*

$$\begin{aligned} c_t = & [1 - \beta(1 - \lambda\chi)] \hat{y}_t - (1 - \lambda) \beta \sigma r_t + \beta(1 - \lambda\chi) E_t c_{t+1} \\ & + \beta \lambda \zeta \left(\chi - \frac{\alpha}{\lambda} \right) (g_t - E_t g_{t+1}) \end{aligned}$$

- Multiplier (fixed- r trick from Bilbiie, 2008, 2011)

	Total Ω	Ind. share ω	Fisc. Mult. $\mathcal{M}$
TANK	$\frac{\sigma}{1-p} \frac{1-\lambda}{1-\lambda\chi}$	$\frac{1-\beta(1-\lambda\chi)}{1-\beta p(1-\lambda\chi)}$	$1 + \frac{\lambda\zeta}{1-\lambda\chi} \left(\chi - \frac{\alpha}{\lambda} \right)$

TANK: Fiscal Multipliers

$$\mathcal{M} = 1 + \frac{\lambda \zeta}{1 - \lambda \chi} \left[\underbrace{(\chi - 1)}_{\text{NK cross}} + \underbrace{\left(1 - \frac{\alpha}{\lambda}\right)}_{\text{exog. redistrib.}} \right] > 1$$

+amplification $\frac{\partial \mathcal{M}}{\partial \lambda} > 0$ w/ uniform t^j , $\alpha = \lambda$ IFF

$$\chi > 1$$

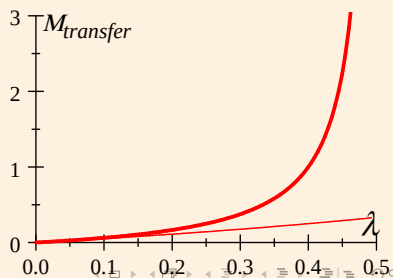
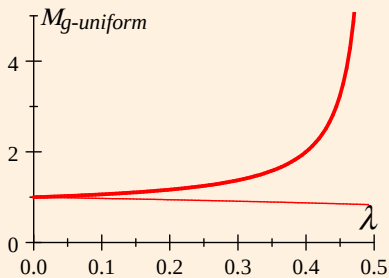
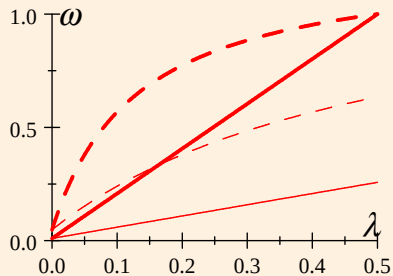
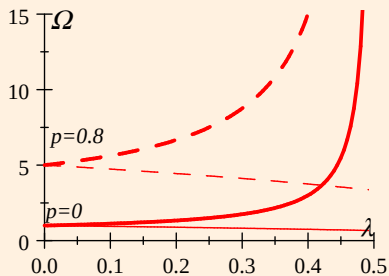
- ▶ $\mathcal{M}$ "total effect": Same decomposition, persistence irrelevant.
 - ▶ $\chi \uparrow$ increases PE slope; also PE shift but *only if* $\chi > 1$
 - ▶ $\alpha \downarrow$ increases PE shift *only if transfer* (progressive shock, $\alpha < \lambda$)
 - ▶ λ increases $\mathcal{M}$; but iff $\chi > 1$ when taxation uniform $\alpha = \lambda$

TANK: Fiscal Multipliers, Previous Work

- ▶ **G spending** multiplier:
 - ▶ numerical, with K (Gali Lopez-Salido Valles JEEA 2007); analytical, no K (dist. taxes) Bilbiie Straub 2004 WP, Bilbiie Meier Mueller 2008 JMCB; Monacelli Perotti IMF EcRev 2012
- ▶ Redistribution (**transfer**) multiplier:
 - ▶ Bilbiie Monacelli Perotti EJ 2013; Mehrotra IJCB; Giambattista and Pennings EER
- ▶ At the **ZLB** with borrower-saver and deleveraging:
 - ▶ Eggertsson Krugman QJE 2012

HANK fiscal multipliers: recent work

- ▶ Hagedorn Manovskii Mitman (AEJ Macro RR): both G and transfers
 - ▶ role of B-rule, deficit financing, "iMPCs" (see later), size of transfer
- ▶ Ferrière Navarro (REStud RR): role of tax progressivity
- ▶ Auclert Rognlie Straub (JPE RR): role of iMPCs for transfer multiplier and deficit-spending G
 - ▶ need iMPCs/liquidity to get positive PDV multipliers w/ fixed- r or $-P$ ($= 0$ in TANK w/ fixed- r or $-P$, see Bilbiie Monacelli Perotti)
- ▶ Cantore Freund (JME): multipliers in analytical model with bonds in utility and iMPCs
- ▶ Kopiec (EER RR): analytical insights in rich HANK



Adding Phillips curve

- ▶ Not much difference, same as in RANK
- ▶ Use $w_t = (\varphi + \sigma^{-1}) c_t + \varphi g_t$

$$\pi_t = \kappa c_t + \kappa \zeta g_t$$

Add Taylor $r_t = i_t - E_t \pi_{t+1} = (\phi - 1) E_t \pi_{t+1} - \varepsilon_t$

- ▶ Replace in Euler-IS-AD

$$c_t = \left[1 - \frac{1 - \lambda}{1 - \lambda \chi} \sigma (\phi - 1) \kappa \right] E_t c_{t+1} + \frac{1 - \lambda}{1 - \lambda \chi} \sigma \varepsilon_t \\ - \underbrace{\frac{1 - \lambda}{1 - \lambda \chi} \sigma (\phi - 1) \kappa \zeta E_t g_{t+1}}_{\text{RANK}} + \underbrace{\frac{\lambda \zeta}{1 - \lambda \chi} \left(\chi - \frac{\alpha}{\lambda} \right)}_{\text{TANK}} (g_t - E_t g_{t+1})$$

Adding Phillips curve

- persistence p

$$c_t = \frac{(1 - \lambda) \sigma}{(1 - \lambda \chi) (1 - p) + p (1 - \lambda) \sigma \kappa (\phi - 1)} \varepsilon_t + \zeta \frac{\lambda (\chi - \frac{\alpha}{\lambda}) (1 - p) - p (1 - \lambda) \sigma \kappa (\phi - 1)}{(1 - \lambda \chi) (1 - p) + p (1 - \lambda) \sigma \kappa (\phi - 1)} g_t$$

- future inflation + active Taylor $\rightarrow$ lower multipliers (intertemporal substitution)
- Note: Liquidity Trap (Zero Lower Bound) multipliers are nested here for $\phi = 0$

What (else) determines χ ?

- ▶ *Fiscal redistribution:*

- ▶ more general progressive taxation (Heathcote Storesletten Violante; Ferrière Navarro; Auclert Rognlie Straub);
- ▶ crucial ingredient in **all** HANK: here spelled out transparently

- ▶ *Sticky wages*

- ▶ TANK: Colciago, w/ Ascari & Rossi; Furlanetto;
- ▶ HANK: Broer et al, Walsh, Auclert Rognlie Straub, Alves Kaplan Moll Violante, Bilbiie Känzig Surico

" χ " as in ... Key(nes)!

$$\chi \equiv 1 + \varphi \left(1 - \frac{\tau^D}{\lambda} \right) \gtrless 1.$$

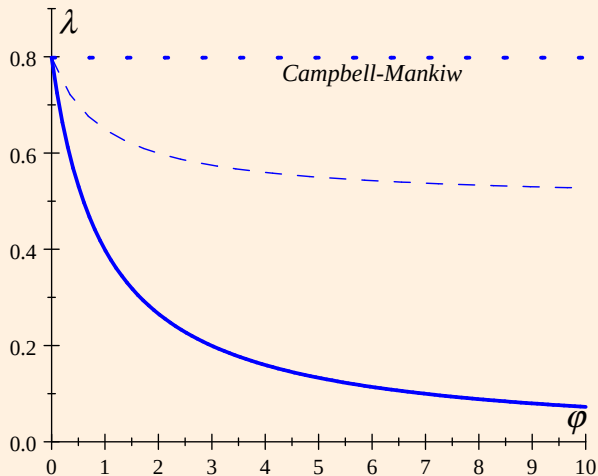
*"The amount that the community spends on **consumption** obviously depends on [...] the principles on which **income is divided** between the individuals composing it (which may suffer **modification** as **output is increased**)."*

*"[...] we may have to make an allowance for the possible **reactions** of **aggregate consumption** to the **change in the distribution** of a given real income ... **resulting from a change in the wage-unit**".*

*"If **fiscal policy** is used as a deliberate instrument for the more equal **distribution of incomes**, its effect in increasing the **propensity to consume** is, of course, all the greater."* Keynes [1936] , Ch. 8, Books I and III

"HANK Surface": Indirect Effect = 0.8

(φ, λ) s.t. $\omega = 0.8, \tau^D = 0.5$ (dash); iid shock



Indirect Amplification

- ▶ Suppose $\mathcal{A}$ times amplification relative to $\lambda = 0$ (RANK):

$$\Omega(\lambda) = \mathcal{A} * \Omega(0)$$

- ▶ *Proposition*: the indirect share is **at least** (for iid shocks):

$$\omega \geq 1 - \frac{1}{\mathcal{A}}$$

- ▶ twice as much effect, at least half of it is indirect; four times, three quarters is indirect, etc.
- ▶ NOTE: invariant to λ and χ

Homework

- ▶ Assume a different fiscal redistribution scheme (your choice) and derive the χ
- ▶ Is the FG puzzle still a puzzle in TANK? More so, less so, or exactly the same as in RANK?

THANK

THANK Model (Ingredients)

- ▶ **Two states:** constrained hand-to-mouth H and unconstrained "savers" S
 - ▶ switch *exogenously* (idiosyncratic uncertainty).
- ▶ Insurance:
 - ▶ *full within* type (after idiosyncratic uncertainty revealed)
 - ▶ *limited across* types.
- ▶ **Two assets** and *liquidity*:
 - ▶ bonds are **liquid** (*can* be used to self-insure, before idiosyncratic uncertainty is revealed)
 - ▶ stocks are **illiquid** (cannot ——— „———).
- ▶ Bond trading
 - ▶ equilibrium liquidity
 - ▶ or **not** (*most analytical* HANK): "**Bondless** limit"

Two-state-, Two-asset, Tractable-HANK

- ▶ shocks $S \leftrightarrow H$
- ▶ $p(S|S) = s; p(H|S) = 1 - s;$
- ▶ $p(H|H) = h; p(S|H) = 1 - h$
- ▶ H mass (unconditional H probability, stationary distribution):

$$\lambda = \frac{1 - s}{2 - s - h}$$

solution of

$$\begin{pmatrix} \lambda & 1 - \lambda \end{pmatrix} \begin{pmatrix} h & 1 - h \\ 1 - s & s \end{pmatrix} = \begin{pmatrix} \lambda & 1 - \lambda \end{pmatrix}$$

Assets

- ▶ S : illiquid shares (profits); liquid nominal public debt
 - ▶ Adjust portfolio *before* knowing if S or H next
 - ▶ If $\rightarrow H$ can only take bonds
- ▶ Bond flows (per capita)
 - ▶ B_{t+1}^S *beginning-of-period- $t+1$, after consumption-saving choice, also after changing state and pooling*
 - ▶ Z_{t+1}^S *end-of-period- t after the consumption-saving choice but before moving*

$$\begin{aligned}(1 - \lambda) B_{t+1}^S &= (1 - \lambda) s Z_{t+1}^S + (1 - \lambda) (1 - s) Z_{t+1}^H \\ \lambda B_{t+1}^H &= (1 - \lambda) (1 - s) Z_{t+1}^S + \lambda h Z_{t+1}^H.\end{aligned}$$

rescaling and using $\lambda = \frac{1-s}{1-s+1-h}$:

$$\begin{aligned}B_{t+1}^S &= s Z_{t+1}^S + (1 - s) Z_{t+1}^H \\ B_{t+1}^H &= (1 - h) Z_{t+1}^S + h Z_{t+1}^H.\end{aligned}$$

Family (Head) Optimization

$$W(B_t^S, B_t^H, \Theta_t) = \max_{\{C_t^S, Z_{t+1}^S, Z_{t+1}^H, C_t^H, \Theta_{t+1}\}} (1 - \lambda) U(C_t^S) + \lambda U(C_t^H) \\ + \beta E_t W(B_{t+1}^S, B_{t+1}^H, \Theta_{t+1})$$

subject to:

$$C_t^S + Z_{t+1}^S + v_t \Theta_{t+1} = Y_t^S + R_t B_t^S + \Theta_t (v_t + D_t), \\ C_t^H + Z_{t+1}^H = Y_t^H + R_t B_t^H \\ Z_{t+1}^S, Z_{t+1}^H \geq 0$$

and the laws of motion for bond flows relating the Z s to the B s

Euler Equations

Look like Bewley-Aiyagari-Huggett...

$$U' \left(C_t^S \right) \geq \beta E_t \left\{ \frac{v_{t+1} + D_{t+1}}{v_t} U' \left(C_{t+1}^S \right) \right\} \text{ and } \Theta_{t+1} = \Theta_t = (1 - \lambda)^{-1};$$

$$U' \left(C_t^S \right) \geq \beta E_t \left\{ R_{t+1} \left[s U' \left(C_{t+1}^S \right) + (1 - s) U' \left(C_{t+1}^H \right) \right] \right\} \text{ and}$$

$$0 = Z_{t+1}^S \left[U' \left(C_t^S \right) - \beta E_t \left\{ R_{t+1} \left[s U' \left(C_{t+1}^S \right) + (1 - s) U' \left(C_{t+1}^H \right) \right] \right\} \right]$$

$$U' \left(C_t^H \right) \geq \beta E_t \left\{ R_{t+1} \left[(1 - h) U' \left(C_{t+1}^S \right) + h U' \left(C_{t+1}^H \right) \right] \right\} \text{ and}$$

$$0 = Z_{t+1}^H \left[U' \left(C_t^H \right) - \beta E_t \left\{ R_{t+1} \left[(1 - h) U' \left(C_{t+1}^S \right) + h U' \left(C_{t+1}^H \right) \right] \right\} \right]$$

Two-state-, Two-asset, Tractable-HANK

- ▶ *Liquidity* (Kaplan et al, ...): $S \xrightarrow{1-s} H$ take **bonds** (liquid), not stock
 - ▶ **self-insurance** (bonds priced even when not traded):

$$\left(C_t^S\right)^{-\frac{1}{\sigma}} = \beta E_t \left\{ (1 + r_t) \left[s \left(C_{t+1}^S\right)^{-\frac{1}{\sigma}} + (1 - s) \left(C_{t+1}^H\right)^{-\frac{1}{\sigma}} \right] \right\}$$

- ▶ "wealthy" H : Euler with inequality (constrained):

$$C_t^H = Y_t^H$$

- ▶ no-liquidity, **bondless** limit: *most analytical* HANK
- ▶ w/ liquidity: $C_t^H + Z_{t+1}^H = Y_t^H + R_t B_t^H$

Step back

- ▶ Designed to capture how ... *which* asset markets work?

Step back

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- ▶ Obviously: ...

Step back

- ▶ Designed to capture how ... *which* asset markets work?
- ▶ Obviously: ...
- ▶ **None!**

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- ▶ Obviously: ...
- ▶ **None!**
- ▶ Designed to mimick equilibrium of (be observationally-equivalent to) micro-consistent model that does describe how certain asset markets work

Step back

- ▶ Designed to capture how ... *which* asset markets work?
- ▶ Obviously: ...
- ▶ **None!**
- ▶ Designed to mimick equilibrium of (be observationally-equivalent to) micro-consistent model that does describe how certain asset markets work
- ▶ Friedman as philosopher of science: approx. "*judge model by implications not (a fortiori unrealistic) assumptions*" (the **F-twist**, cf Samuelson)

Income Inequality and Risk: Cyclicalty

- **Income inequality** ($\sim$ Gini, generalized entropy)

$$\Gamma_t(Y_t) \equiv \frac{Y_t^S}{Y_t^H}$$

- **"Risk"** (Ravn Sterk; Challe Matheron Ragot Rubio; Werning; Acharya Dogra)

$1 - s(Y_{t+1}), -s_Y \gtrless 0 \rightarrow$ pro-(counter-)cyclical risk (NB: λ invariant)

Conditional variance Storesletten Telmer Yaron & **skewness** Guvenen Ozkan Song

$$\begin{aligned} \text{var}(\ln Y_{t+1}^S | \ln Y_t^S) &= s(.) (1 - s(.)) (\ln \Gamma_{t+1})^2 \\ \rightarrow \frac{d(\text{var})}{dY} &= \frac{1-s}{Y} \left(\underbrace{\frac{-s_Y Y}{1-s} (2s-1) (\ln \Gamma)^2}_{\text{pure risk, skewness}} + \underbrace{\frac{\Gamma_Y Y}{\Gamma} s \ln(\Gamma)^2}_{\text{inequality}} \right) \end{aligned}$$

- $s = 1$: TANK – no risk
- $s = 0$: oscillating, $\lambda = 1/2$ – no risk (Woodford '90)
- approximate $\sim$ SS $Y^S = Y^H \rightarrow \Gamma = 1 \rightarrow$ risk 1st-order acyclical

Rest: TANK

- Recall loglin (around $D = 0$) equilibrium:

$$\begin{aligned}c_t^H &= \chi y_t = \left[1 + \underbrace{\varphi}_{\text{labor mkt.}} \times \underbrace{\left(1 - \frac{\tau^D}{\lambda}\right)}_{\text{fiscal redistrib.}} \right] y_t \\c_t^S &= \frac{1 - \lambda\chi}{1 - \lambda} y_t\end{aligned}$$

Extra income effect $w \uparrow \rightarrow d \downarrow$ *keystone*: profits

$$\text{Cyclical Income Inequality: } \gamma_t = y_t^S - y_t^H = \frac{1 - \chi}{1 - \lambda} y_t$$

- (χ) : Any income-distribution model

Aggregate Euler in THANK

- Aggregate, replace $c_t^j \longrightarrow c_t^S = sE_t c_{t+1}^S + (1-s)E_t c_{t+1}^H - \sigma r_t$

$$c_t = \underbrace{\left[1 + (\chi - 1) \frac{1-s}{1-\lambda\chi} \right]}_{\equiv \delta} E_t c_{t+1} - \sigma \underbrace{\frac{1-\lambda}{1-\lambda\chi}}_{\text{TANK}} r_t$$

Aggregate Euler in THANK

$$c_t = \underbrace{\left[1 + (\chi - 1) \frac{1-s}{1-\lambda\chi} \right]}_{\equiv \delta} E_t c_{t+1} - \underbrace{\sigma \frac{1-\lambda}{1-\lambda\chi}}_{\text{TANK}} r_t$$

1. **TANK** Amplification iff $\chi > 1$: **Inequality** Countercyclical

Generalizes to rich-HANK: $\text{cov}(\text{MPC}, \chi)$, Auclert JMP 2015; Micro evidence: Patterson JMP 2019

$$\text{agg. MPC} \equiv \lambda \times 1 \times \chi + (1 - \lambda) \times (1 - \beta) \times \frac{1 - \lambda\chi}{1 - \lambda}$$

Aggregate Euler in THANK

$$c_t = \underbrace{\left[1 + (\chi - 1) \frac{1 - s}{1 - \lambda \chi} \right]}_{\equiv \delta} E_t c_{t+1} - \sigma \underbrace{\frac{1 - \lambda}{1 - \lambda \chi}}_{\text{TANK}} r_t$$

1. **TANK Amplification** iff $\chi > 1$: **Inequality Countercyclical**

Generalizes to rich-HANK: $\text{cov}(\text{MPC}, \chi)$, Auclert JMP 2015; Micro evidence: Patterson JMP 2019

$$\text{agg. MPC} \equiv \lambda \times 1 \times \chi + (1 - \lambda) \times (1 - \beta) \times \frac{1 - \lambda \chi}{1 - \lambda}$$

2. **THANK Compounding/Discounting** $\delta \gtrless 1$ iff $\chi \gtrless 1$

- ▶ same as in **TANK** but intertemporal! (amplification to news)
- ▶ *Not necessarily* cyclical risk

THANK Amplification: Acyclical Risk

$$c_t = \underbrace{\left[1 + (\chi - 1) \frac{1-s}{1-\lambda\chi} \right]}_{\equiv \delta} E_t c_{t+1} - \sigma \underbrace{\frac{1-\lambda}{1-\lambda\chi}}_{\text{TANK}} r_t$$

- ▶ *Not necessarily* cyclical risk:
- ▶ (i) $\sim Y^H = Y^S \rightarrow \Gamma = 1 \rightarrow$ variance is zero to first order
- ▶ (ii) $s = 0$ (Woodford 1990) $\lambda = 1/2$ agents oscilate, Aggregate Euler:

$$c_t = \frac{\chi}{2-\chi} E_t c_{t+1} - \sigma \frac{1}{2-\chi} r_t$$

$$\delta|_{s=0} = \frac{\chi}{2-\chi} \leq 1 \text{ iff } \chi \leq 1.$$

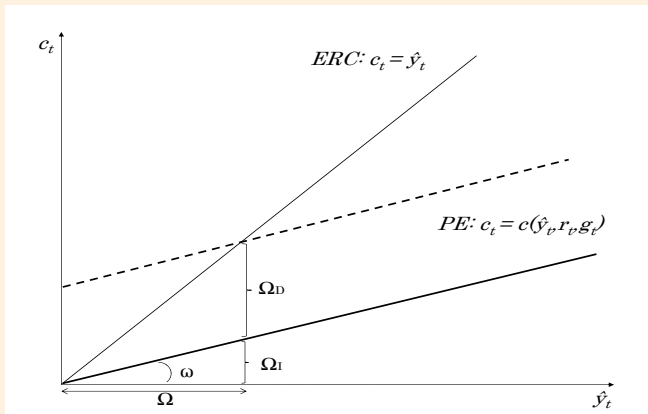
The New Keynesian Cross in THANK

$$\text{PE: } c_t = [1 - \beta (1 - \lambda\chi)] y_t - (1 - \lambda) \beta \sigma r_t + \beta \delta (1 - \lambda\chi) E_t c_{t+1}$$

	Total effect Ω ("multiplier")	Indirect-effect share ω ("aggregate MPC")
TANK	$\frac{\sigma}{1-p} \frac{1-\lambda}{1-\lambda\chi}$	$\frac{1-\beta(1-\lambda\chi)}{1-\beta p(1-\lambda\chi)}$
THANK	$\frac{\sigma}{1-\delta p} \frac{1-\lambda}{1-\lambda\chi}$	$\frac{1-\beta(1-\lambda\chi)}{1-\delta\beta p(1-\lambda\chi)}$

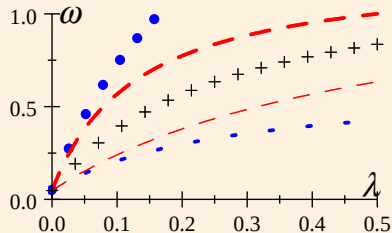
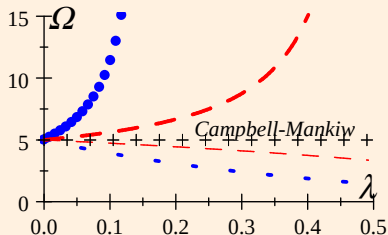
The New Kenesian Cross

$$c_t = \omega \hat{y}_t - (1 - \omega) \Omega r_t + (1 - \omega) (M - 1) g_t$$



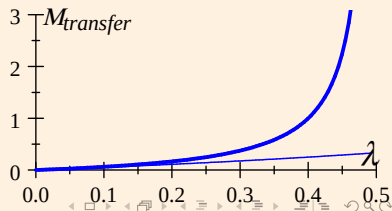
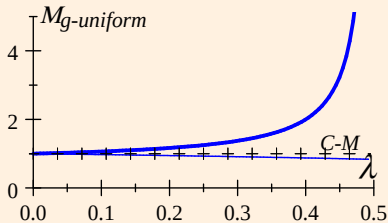
$$\text{agg. MPC } \omega \equiv \lambda \times 1 \times \chi + (1 - \lambda) \times (1 - \beta) \times \frac{1 - \lambda \chi}{1 - \lambda}$$

The New Kenesian Cross (in THANK)



Calibrate **KMV** $\Omega/\Omega_{rank} = 1.5$, $\omega = .8$:

$\chi = 1.42$; $\lambda = .37$; $1 - s = .04$



Calibrating Simple to Match Complicated

Table 1: Approximating HANK

HANK: Equilibrium objects					Implied parameters		
	$\frac{\Omega}{\Omega^*}$	ω	$\frac{\Omega_1^F}{\Omega^*}$	$\frac{\Omega_{20}^F}{\Omega^*}$	χ	λ	$1 - s$
Kaplan et al	1.5	.8	—	—	1.48 1.48	.41 .37	0 (TANK) .04
McKay et al	—	—	.8	.4	— .3	— .21	0 (TANK) .04

Paper: other HANKs (Goremann et al, Debortoli Gali, Hagedorn et al, Auclert et al)

A Different Cyclical-Risk Channel

- Aggreg. Euler w/ $-s'(Y_{t+1}) \geq 0$ ($\Gamma = Y^S/Y^H > 1$):

Ravn Sterk; Werning; Acharya Dogra

$$c_t = (\delta + \eta) E_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} r_t$$

$$\eta \equiv \frac{s_Y Y}{1-s} \left(1 - \Gamma^{-1/\sigma}\right) (1 - \tilde{s}) \sigma \frac{1-\lambda}{1-\lambda\chi}$$

- ▶ **Similar** equilibrium Euler-discounting/*compounding*
- ▶ *Different* precautionary saving: prudence $\sigma > 0$
- ▶ "As if" $\tilde{\delta} \equiv \delta + \eta$ (Carroll Kimball Ecma 96)
- ▶ Reduced-form $\Leftrightarrow$ Acharya Dogra PRANK (discounting, no contemp. multipliers)
- ▶ different interpretation: PRANK=CARA+normality $\rightarrow$ cyclical variance, no skewness; **here**: $\eta \sim$ cyclicality of skewness, variance cyclical for two reasons

A Common Misconception: "Constant" Euler Wedges?

- ▶ No, **not constant**: they are **cyclical** (that is the whole point)
- ▶ their *elasticities* (to some *endogenous* variables) are constant around the long-run ergodic steady-state

$$\begin{aligned}
 c_t = & \underbrace{E_t c_{t+1} - \sigma r_t}_{\text{RANK}} - \underbrace{\sigma \frac{\lambda(\chi - 1)}{1 - \lambda\chi} r_t}_{\text{cyc.-ineq. TANK}} \\
 & + \underbrace{(\delta - 1) E_t c_{t+1}}_{\text{cyc.-ineq.+risk THANK}} + \underbrace{\eta E_t c_{t+1}}_{\text{(pure) cyc.-risk THANK}}
 \end{aligned}$$

- ▶ Debortoli Galí, formal comparison of (Bilbiie 2008) TANK w/ a quantitative HANK, good approx. for some shocks less for others.
- ▶ analytical version of their decomposition: "between" (TANK) vs "within" (second line)
- ▶ see also Bocola et al "accounting/wedges"

The 3-Equation THANK Model

$$c_t = \delta E_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} (i_t - E_t \pi_{t+1})$$

$$: \quad (\text{with } \delta \equiv 1 + (\chi - 1) \frac{1 - s}{1 - \lambda \chi})$$

$$\pi_t = \kappa c_t + \beta E_t \pi_{t+1}$$

$$i_t = \phi \pi_t$$

-
- (here $\pi_t = \kappa c_t$ simple closed forms, paper NKPC)

The 1-Equation THANK Model

$$\begin{aligned}c_t &= \delta E_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} (i_t - E_t \pi_{t+1}) \\&: \quad (\text{with } \delta \equiv 1 + (\chi - 1) \frac{1 - s}{1 - \lambda \chi}) \\ \pi_t &= \kappa c_t \\ i_t &= \phi \pi_t\end{aligned}$$

-
- (here $\pi_t = \kappa c_t$ simple closed forms, paper NKPC)

The HANK Taylor Principle

$$c_t = \frac{\delta + \kappa\sigma \frac{1-\lambda}{1-\lambda\chi}}{1 + \phi\kappa\sigma \frac{1-\lambda}{1-\lambda\chi}} E_t c_{t+1} + \text{shocks}$$

- ▶ $\exists!$ REE (local determinacy) with $\lambda < \chi^{-1}$:

$$\phi > 1 + \frac{\delta - 1}{\kappa\sigma \frac{1-\lambda}{1-\lambda\chi}}.$$

- ▶ **Taylor principle** $\phi > 1$ *sufficient* if:

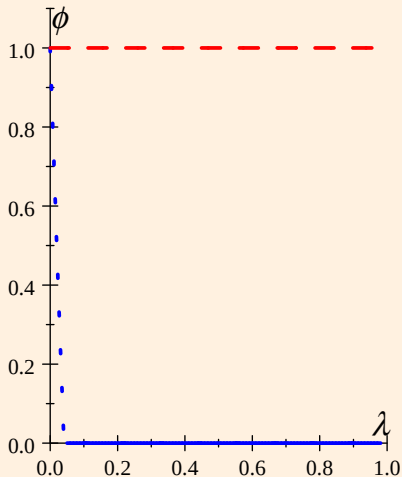
$$\delta \leq 1 \longrightarrow \chi \leq 1 \quad (\Leftrightarrow \Sigma(\text{iMPCs}) \geq 1, \text{ Auclert Rognlie Straub—see later})$$

- ▶ subsequently: Acharya Dogra (Ecma 2020) w/ **cyclical (pure) risk**

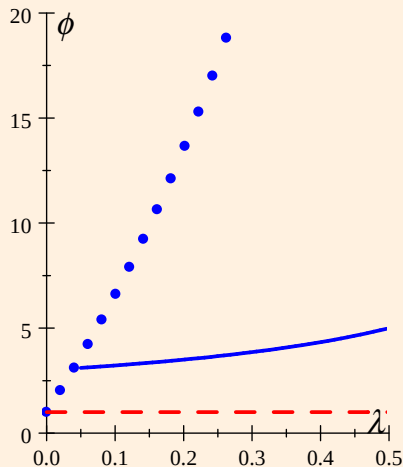
$$\phi > 1 + \frac{\delta - 1 + \eta}{\kappa\sigma \frac{1-\lambda}{1-\lambda\chi}}.$$

The HANK Taylor Principle and Sargent-Wallace

Threshold ϕ : TANK (dash); $s = .96$ (solid); iid: $1 - s = \lambda$ (dots)



PI: $\chi < 1$



CI: $\chi > 1$

Catch-22: No Puzzle, No Amplification?

1. HANK Amplification-Multiplier iff:

$$\chi > 1$$

intuition: *NK Cross*; **paper**: liquidity traps, fiscal multipliers

2. **No-puzzle** iff *HANK-Disc.* > *RANK-Comp.*

$$v_0 = \delta + \kappa\sigma \frac{1-\lambda}{1-\lambda\chi} < 1 \longrightarrow \chi < < 1$$

Proof: $c_t = v_0 E_t c_{t+1} - \sigma \frac{1-\lambda}{1-\lambda\chi} i_t^* = v_0^{\bar{T}} E_t c_{t+\bar{T}} - \sigma \frac{1-\lambda}{1-\lambda\chi} E_t \sum_{j=0}^{\bar{T}-1} v_0^j i_{t+j}^*$

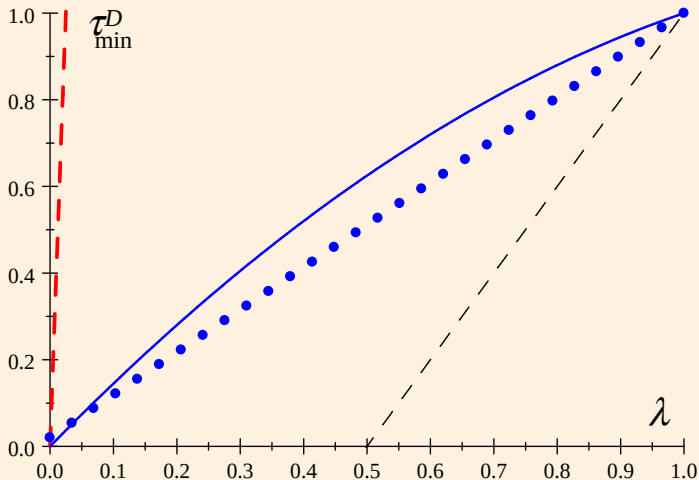
FG Puzzle: Resolved or Aggravated

- ▶ **Aggravated** with countercyclical inequality $\chi > 1$
- ▶ Also: discounting $\delta < 1$ not sufficient; sufficiency:

$$1 - s > 0 \text{ and } \chi < 1 - \sigma\kappa \frac{1 - \lambda}{1 - s} < 1$$

- ▶ **McKay Nakamura Steinsson**: sufficient conditions for resolving FG puzzle, two special cases
 - ▶ analytical, $\chi = 0$, $\delta = s$, iid $s = 1 - \lambda$
 - ▶ quantitative: rebate profits uniformly, i.e. disproportionately more to bottom ("poor"), isomorphic to $\tau^D > \lambda$ so $\chi < 1$
- ▶ **Hagedorn Luo Manovskii Mitman**: more quantitative examples of both cases (sticky wages, redistribution, etc.)

No-Puzzle Threshold Redistribution



Redistribution threshold $\tau_{\min}^D$ in TANK 1 – $s \rightarrow 0$ (dash);
1 – $s = 0.04$ (solid); iid HANK 1 – $s = \lambda$ (dots).

Cyclical Inequality vs Risk: Solution to Catch-22?

- ▶ Yes and No
- ▶ *No-Catch-22*: **Amplification without Puzzles** iff

Countercyclical Inequality : $\chi \geq 1$

Procyclical (enough) Risk : $\eta < 1 - \delta < 0$

- ▶ Flip side: *everything worse* if (pure) **risk countercyclical too**
 $\eta > 0$
-

- ▶ **Catch-22** (like the FG puzzle!): **Conditional** on policy rule
....
- ▶ Indeterminacy w/ Taylor pervasive w/ countercyclical inequality +++ countercyclical risk

Virtuous Policies in HANK 1: Wicksellian PLT Rule

- ▶ = *Amplification, Determinacy & No Puzzle*
- ▶ **Wicksellian** price-level-targeting: $\exists!$ REE w/

$$i_t = \phi_p p_t \text{ with } \phi_p > 0 \text{ (Woodford \& Giannoni in RANK)}$$

- ▶ Model:

$$c_t = v_0 E_t c_{t+1} - \sigma \frac{1-\lambda}{1-\lambda\chi} \phi_p p_t; v_0 \equiv \delta + \kappa \sigma \frac{1-\lambda}{1-\lambda\chi}$$

$$\text{PC} : p_t - p_{t-1} = \kappa c_t.$$

$$\rightarrow E_t p_{t+1} - \left[1 + v_0^{-1} \left(1 + \sigma \frac{1-\lambda}{1-\lambda\chi} \phi_p \kappa \right) \right] p_t + v_0^{-1} p_{t-1} = 0$$

- ▶ Intuition: **PID control**—bygones *not* bygones;³

³Corrolary: *also* in **RANK**!

THANK with Bonds-Liquidity

THANK with bonds-liquidity

- Individual BC w/ liquidity

$$C_t^H + Z_{t+1}^H = \hat{Y}_t^H + R_t B_t^H$$

asset-market equilibrium:

$$Z_{t+1}^H = 0(\text{impatient}) \longrightarrow B_{t+1} = (1 - \lambda) Z_{t+1}^S;$$

$$B_{t+1}^H = \underset{(\frac{(1-\lambda)(1-s)}{\lambda})}{(1-h)} Z_{t+1}^S = \frac{1-h}{1-\lambda} B_{t+1} = \frac{1-s}{\lambda} B_{t+1}$$

$$B_{t+1}^S = s Z_{t+1}^S = \frac{s}{1-\lambda} B_{t+1}$$

- Replace in indiv BCs

$$C_t^S + \frac{1}{1-\lambda} B_{t+1} = \hat{Y}_t^S + \frac{s}{1-\lambda} R_t B_t$$

$$C_t^H = \hat{Y}_t^H + R_t \frac{1-s}{\lambda} B_t$$

- Euler self-insurance $\rightarrow$ savings (liquidity) demand

Demand for Bonds-Liquidity in THANK

- SS: use income formulae and balanced-budget $T = rB$ (normalize $Y = 1$), log utility $\sigma = 1$:

$$B = \frac{\frac{1}{1 + \left(\frac{1}{\beta R} - 1\right)^{\frac{1-\lambda}{1-s}}} - \frac{1}{\mu}}{1 + R \left(\frac{1-s}{\lambda} - 1\right)}$$

- Note

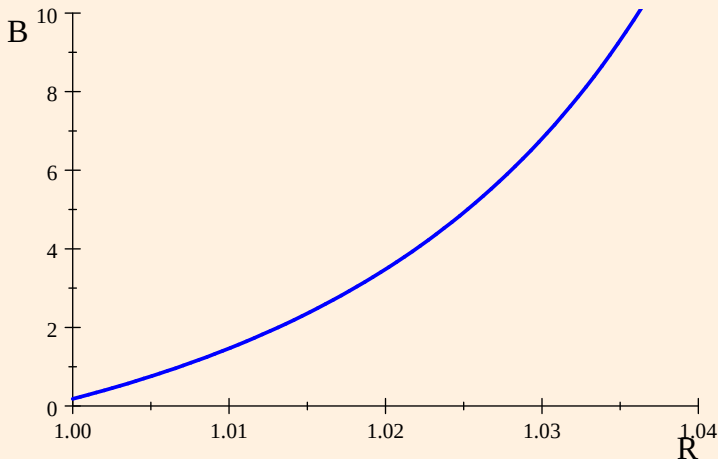
$$B \xrightarrow{R \rightarrow \beta^{-1}} \frac{1 - \mu^{-1}}{1 + \beta^{-1} \left(\frac{1-s}{\lambda} - 1\right)}$$

$\rightarrow 0$ when $\mu = 1$ and $B > 0$ iff

$$\frac{1-s}{\lambda} > 1 - \beta$$

(more general restrictions on $1 - s$, λ , β to have positive demand)

SS Bonds/Liquidity Demand (Savings Function)



$$(1 - s, \lambda, \beta, \mu) = (.04, .33, .99, 1.2)$$

iMPCs in THANK w/ liquidity

- ▶ **Auclert Rognlie Straub**; Hagedorn Manovskii Mitman
 - ▶ *fiscal policy*
- ▶ most compelling critique of **TANK** ... **not of THANK!**
- ▶ better still: χ helps match data (Fagereng Holm Natvik)

iMPCs in THANK (w/ liquidity)

- ▶ loglin. indiv. BCs, replace in self-insurance Euler $\rightarrow$ demand for liquidity
- ▶ at given income (no govt BC): take partial derivative wrt aggregate income shock, keeping fixed everything
- ▶ Special oscillating case: $s = 0$ and $\lambda = \frac{1}{2}$. Asset accumulation eq.

$$b_{t+1} = \frac{\hat{y}_t^S - E_t \hat{y}_{t+1}^H}{2(1 + \beta^{-1})} = \frac{1}{2(1 + \beta^{-1})} [(2 - \chi) \hat{y}_t - \chi E_t \hat{y}_{t+1}]$$

(dis-)save when expect (higher) lower income tomorrow.

- ▶ Consumption function

$$c_t = \frac{2 - \chi + \beta\chi}{2(1 + \beta)} \hat{y}_t + \frac{2 - \chi}{2(1 + \beta)} \hat{y}_{t-1} + \frac{\beta\chi}{2(1 + \beta)} \hat{y}_{t+1}$$

iMPCs in THANK (w/ liquidity)

- Proposition: iMPCs in oscillating model $s = 0$

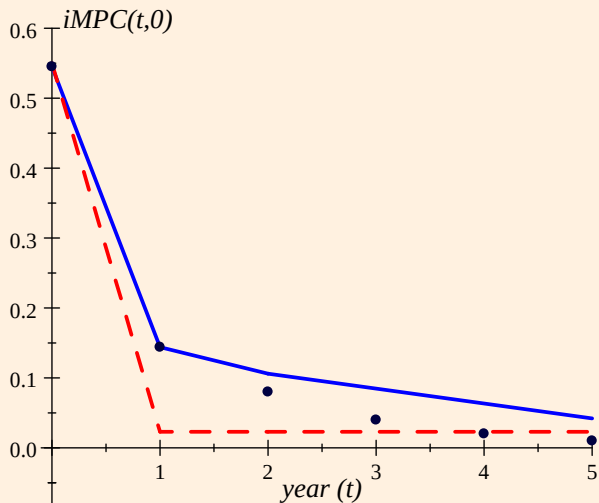
$$\frac{dc_T}{d\hat{y}_T} = \frac{2 - \chi + \beta\chi}{2(1 + \beta)}; \quad \frac{dc_{T+1}}{d\hat{y}_T} = \frac{2 - \chi}{2(1 + \beta)}; \quad \frac{dc_{T-1}}{d\hat{y}_T} = \frac{\beta\chi}{2(1 + \beta)}$$

$$\frac{dc_t}{d\hat{y}_T} = 0 \text{ o/w}$$

General proposition: paper (closed-form but unwidely)

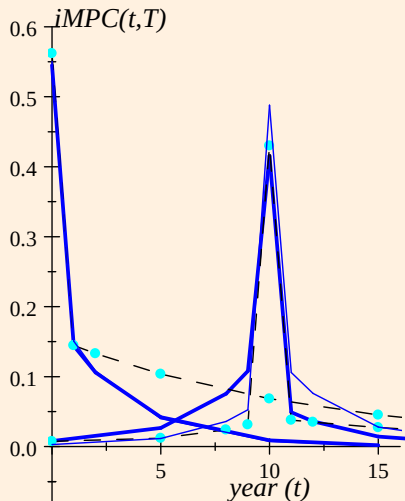
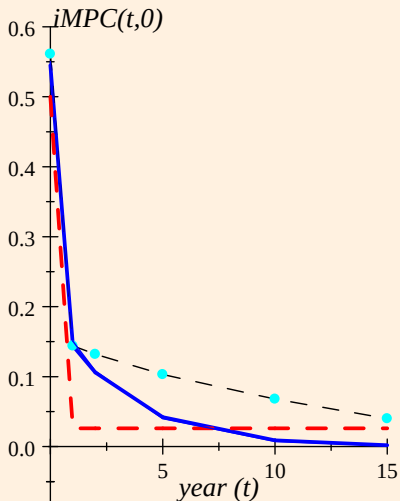
- Alternative analytics for iMPCs: TANK & bonds in utility w/ adjustment costs, Cantore Freund 2021

iMPCs in THANK



iMPCs in THANK (blue solid); TANK (red dash); Data (dots)

iMPCs in THANK



$\chi = 1$ (dot-dash); TANK (red dash); $\chi > 1$ thick and < 1 thin

Virtuous Policies in HANK 2: Debt Quantity Rule

- ▶ = *Amplification, Determinacy & No Puzzle*
- ▶ Application of **Hagedorn's** result/idea: (Nominal) **Debt-quantity rule** in version w/ liquidity

$$b_{t+1}^N \equiv b_{t+1} + p_t = 0 \text{ or } \phi_b p_t$$

- ▶ Loglin $\sim B = 0$ & $R = \beta^{-1}$ ($\mu = 1$) SS, "Ricardian" FP

$$c_t = \delta E_t c_{t+1} - \sigma \frac{1-\lambda}{1-\lambda\chi} r_t + \frac{1-\lambda-s}{1-\lambda\chi} b_t \\ + \left[\frac{\lambda}{1-\lambda\chi} \beta + \frac{1-\lambda-s}{1-\lambda\chi} \left(\frac{1-s}{\lambda} - 1 \right) \right] b_{t+1} + \frac{1-\lambda-s}{1-\lambda\chi} \beta E_t b_{t+2}$$

- ▶ Add again $c_t = \kappa^{-1} (p_t - p_{t-1})$ determinacy condition(!):

$$\frac{1-s}{\lambda} > 1 - \beta.$$

→ **always determinate** w/ well-defined bonds demand

- ▶ Intuition: ... "**money**" rule

Income Processes ("External Validation")

- ▶ **Income inequality** ($\sim$ Gini, entropy, etc.) $\Gamma_t(Y_t) \equiv Y_t^S / Y_t^H$
- ▶ **Conditional variance** ($\sim$ *income risk*):

$$\text{var} \left(\ln Y_{t+1}^S \mid \ln Y_t^S \right) = s(1-s) (\ln \Gamma_{t+1})^2.$$

- ▶ **Conditional skewness and kurtosis:**

$$\begin{aligned} \text{skew} \left(\ln Y_{t+1}^S \mid \ln Y_t^S \right) &= \frac{1-2s}{\sqrt{s(1-s)}}; \\ \text{kurt} \left(\ln Y_{t+1}^S \mid \ln Y_t^S \right) &= \frac{1}{s(1-s)} - 3 \end{aligned}$$

- ▶ **Autocorrelation** (> 0 if $s \geq 1-h$)

$$\text{corr} \left(\ln Y_{t+1}^j, \ln Y_t^j \right) = s + h - 1 = 1 - \frac{1-s}{\lambda};$$

(Basically Rouwenhorst with 2 states.)

Recap: THANK as HANK projection

1. Idiosyncratic income uncertainty (variance, skewness, kurtosis)
2. NK Cross, *cyclical inequality* χ
3. Self-insurance, precautionary saving: from both constraints & prudence
4. $\rightarrow$ Cyclical income risk (two sources)
5. Bonds/Liquidity and iK Cross: iMPCs

$\rightarrow$ We are **reassured** that we derived loads of analytical results (and will momentarily derive more, e.g. **optimal policy**) in a **relevant model**

Physical Capital - Illiquid Wealth

Based on Bilbiie, Känzig, and Surico:

*Capital and Income Inequality: An
Aggregate-Demand Complementarity*

Motivation

- ▶ Investment in Physical Capital (**K**) central to macro:
 - ▶ growth, inequality, optimal taxation, business cycles
- ▶ Real Business Cycles (RBC): *General Equilibrium* model
 - ▶ K investment ~ 4 times more volatile than GDP, etc.
- ▶ *New Keynesian* (NK) ... **No K**
 - ▶ side show for monetary transmission;
 - ▶ ingredient in DSGE versions, but *back seat* for **Consumption**
Caveat: "financial accelerator" notwithstanding
- ▶ Burgeoning **Heterogeneous-Agent** NK literature (TANK & HANK)
 - ▶ *reappraise the role of K?*

Our Result(s)

- ▶ Heterogeneity puts K front and center in NK!
- ▶ Isolate **two amplification channels** for monetary policy
 - ▶ **cyclical income (Y) inequality** (Bilbiie 2008, ...)
 - ▶ **unequal capital expenditures** (this paper) $\sim$ wealth inequality
- ▶ They are **complementary**!

Amplification of Mon. Pol. Effects on C		
No Y inequality		Y inequality
No K	= repres.-agent	small
K	small	LARGE

$\equiv$ joint effect much larger than individual effects combined

Our Result(s)

- ▶ Interaction w/ fiscal redistribution: essential *what K income*
 - ▶ **opposite effects** of redistributing physical K vs monopoly profits
- ▶ Complementarity robust to idiosyncratic risk, sticky wages, etc.
- ▶ $T_{(\text{ractable})}\text{-HANK w/ } K \rightarrow \underline{\text{analytics}}$ (novel even for RANK)

*ANK: Heterogeneity/Inequality Matters

- ▶ **Empirical:** Campbell Mankiw; Hall; ...; Kaplan Violante; Cloyne Ferreira Surico
- ▶ **RANK+K:** Dupor, Carlstrom Fuerst, Sveen Weinke, Woodford, Rupert Sustek...
- ▶ **2000s TANK:** Galí Lopez-Salido Vallés; Bilbiie...; **borrower-saver:** Iacoviello; Nistico; Eggertsson
Krugman ...
- ▶ **2010s HANK focus on C:** Auclert; Kaplan, Moll, Violante; Gornemann Kuester
Nakajima; McKay Nakamura Steinsson; Guerrieri Lorenzoni; Bayer Luetticke Pham-Dao Tjaden; Auclert
Rognlie; Debortoli Galí; Ravn Sterk; Den Haan Rendahl Riegler; Luetticke; McKay Reis; Challe Matheron
Ragot Rubio; Oh Reis; Hagedorn Manovskii Mitman; Auclert Rognlie Straub; Acharya Dogra; Werning;
Cantore Freund; Ferrante Paustian; Bilbiie Ragot; Cui Sterk; Bhandari Evans Golosov Sargent; Bilbiie; ...
- ▶ now: **HANK with focus on K**
 - ▶ Auclert Rognlie Straub; Alves Kaplan Moll Violante; this paper ...

A Tale of Two Inequalities

- **This paper:**

$$C^j + S^j = (WN)^j + (rK + D)^j + T^j \equiv Y^j$$

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- no K , Income (Y) inequality only: RHS

$$C^j = (WN)^j + D^j + T^j \equiv Y^j$$

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- Bilbiie 2008 JET ("TANK"), many HA models

A Tale of Two Inequalities

- ▶ **This paper:**

$$C^j + S^j = (WN)^j + (rK + D)^j + T^j \equiv Y^j$$

- ▶ no K , Income (Y) inequality only: RHS

$$C^j = (WN)^j + D^j + T^j \equiv Y^j$$

- ▶ Bilbiie 2008 JET ("TANK"), many HA models

- ▶ K , no Y inequality (but still MPC heterogeneity): LHS

$$C^j + S^j = Y$$

Saver-spender: Gali Lopez-Salido Valles 2007, Mankiw 2000, any HA model with *assets in positive net supply*

A Simple Heterogeneous-Agent Model

- λ hand-to-mouth H , $1 - \lambda$ savers S w/ bonds Euler (loglin)

$$c_t^S = E_t c_{t+1}^S - r_t$$

+ isoelastic investment

$$i_t = \eta y_t$$

+ (Loglin) individual BCs ($X_Y \equiv X/Y$ in steady state):

$$\begin{aligned} C_Y c_t^S + \frac{I_Y}{1 - \lambda} i_t &= Y_Y^S y_t^S \\ c_t^H &= y_t^H = \chi y_t, \end{aligned}$$

the key $\chi = a$ model of the income distribution.

Isolating Income (Y) Inequality

- ▶ no K ($I_Y = 0$), all RHS: **Bilbiie 2008→2019**

$$c_t^H = \chi y_t; c_t^S = \frac{1 - \lambda \chi}{1 - \lambda} y_t.$$

- ▶ Aggregate Euler $\sim$ Aggregate Demand

$$c_t = E_t c_{t+1} - \underbrace{\frac{1 - \lambda}{1 - \lambda \chi}}_{\text{multiplier}} r_t$$

amplification iff $\chi > 1$ (countercyclical Y ineq.)

full-HANK generalization Auclert JMP: earnings

micro evidence: **Patterson**; Slacalek et al; Alves et al; Heathcote et al 10; **macro: Bilbiie Primiceri**

Tambalotti $\chi \approx 1.33$

- ▶ **New Keynesian Cross:**

Aggregate MPC(ish): $\lambda \times 1 \times \chi + \dots$

$\sim$ "indirect effect" Kaplan Moll Violante

Isolating K Inequality (LHS): This Paper

- ▶ Assume that income is perfectly redistributed $\chi = 1$:

$$c_t^H = y_t$$

$$C_Y c_t^S + \frac{I_Y}{1 - \lambda} i_t = Y_Y^S y_t.$$

- ▶ Aggregate Euler - Demand:

$$c_t = E_t c_{t+1} - \underbrace{\frac{1 - \lambda}{1 - \lambda \frac{1 - I_Y}{1 - \eta I_Y}}}_{\text{multiplier}} r_t$$

- ▶ Another Keynesian-cross multiplier iff $\eta > 1$ (dah):

the savings rate (of S) acts as an MPC (of H)

S's saving-investment $\rightarrow$ K **income, redistribution** $\rightarrow$ H, **not** saving

- ▶ Novel analytical isolation $\rightarrow$ any HA w/ *some* K (net saving);
- ▶ formalizes Samuleson's famous 1939 multiplier-accelerator

The Multiplier ... of the Multiplier

- ▶ both K and Y inequality

$$\left| \frac{\partial c_t}{\partial r_t} \right| = \frac{1 - \lambda}{1 - \lambda \chi^{\frac{1 - I_Y}{1 - \eta I_Y}}}$$

- ▶ **Complementarity** if Y ineq. countercyclical $\chi > 1$ and investment share procyclical $\eta > 1$:

$$\left| \frac{\partial c_t}{\partial r_t} \right|_{K, Y \text{ ineq}} > \left| \frac{\partial c_t}{\partial r_t} \right|_{no K, Y \text{ ineq}} \times \left| \frac{\partial c_t}{\partial r_t} \right|_{K, no Y \text{ ineq.}}$$

The Multiplier (Samuelson 1948)

$$\frac{1}{1 - x}$$

x (loosely) "aggregate MPC"

The Multiplier ... of the Multiplier

- ▶ Someone's saving/investment is (capital) **income**
- ▶ partly received by someone who is **not** saving ...
 - ▶ *labor income, fiscal redistribution, etc.*

The Multiplier ... of the Multiplier

- ▶ abstract from direct effect scaled down $(1 - \lambda)$, multiplier:

$$\frac{1}{1 - \lambda \chi \frac{1 - I_Y}{1 - \eta I_Y}}$$

- ▶ two indirect effects interact non-linearly at each round, multiplying each other

A picture worth $1/(1-x)$ words

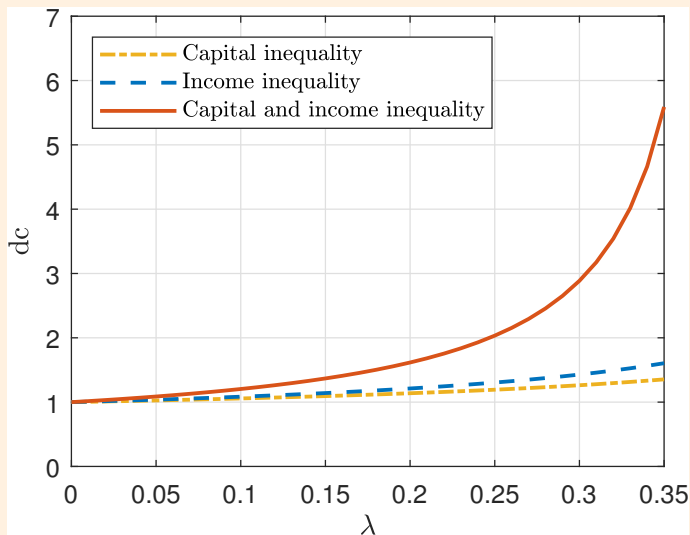


Figure: C multipliers as a function of λ ($\alpha = 0.33, \beta = 0.99, \chi = 1.7$).

Simple Testable Predictions

1. Y and C inequality:

$$\begin{aligned}y_t^S - y_t^H &= \frac{1 - \chi}{(1 - \lambda)Y_Y^S} y_t \\c_t^S - c_t^H &= \frac{1 - \chi C_Y - \eta I_Y}{(1 - \lambda)C_Y} y_t.\end{aligned}$$

both counter-cyclical iff $C_Y (\chi - 1) + \eta I_Y (\eta - 1) > 0$

2. C inequality more counter-cyclical than Y inequality if $\eta > 1$, investment more than 1-to-1 **procyclical**

- compare to evidence (Coibion Gorodnichenko Kueng Silvia 2017)

A Tractable HANK (THANK) with K: This Paper

- ▶ extend Bilbiie (2019) with **illiquid K** (Kaplan Moll Violante, etc.)
- ▶ idiosyncratic uncertainty: two states $S \overset{1-h}{\underset{1-s}{\rightleftharpoons}} H$

$$\lambda = \frac{1-s}{2-s-h}$$

- ▶ assets: *liquid* bonds, *illiquid* K & shares (claims to profits)
- ▶ labor union (sticky wages later)
- ▶ the government redistributes K and/or profits
- ▶ *Reminder*: THANK matches micro moments: iMPCs (Auclert Rognlie Straub), income risk, leptokurtosis, left-skewness, cyclical skewness, etc.

A Tractable HANK (THANK) with K: This Paper

- ▶ Illiquid K ($Q_t \equiv (\Phi'(.))^{-1}$ Tobin's marginal Q)

$$Q_t = \beta E_t \left\{ \left(\frac{C_{t+1}^S}{C_t^S} \right)^{-\frac{1}{\sigma}} \left[(1 - \tau^K) R_{t+1}^K + Q_{t+1} \left(1 - \delta + \Phi_{t+1} - \frac{I_{t+1}}{K_{t+1}} \Phi'_{t+1} \right) \right] \right\}$$

$$K_{t+1} = (1 - \delta) K_t + \Phi \left(\frac{I_t}{K_t} \right) K_t$$

- ▶ Liquid bonds

$$(C_t^S)^{-\frac{1}{\sigma}} = \beta E_t \left\{ \frac{1 + r_t^n}{1 + \pi_{t+1}} \left[s (C_{t+1}^S)^{-\frac{1}{\sigma}} + (1 - s) (C_{t+1}^H)^{-\frac{1}{\sigma}} \right] \right\}$$

- ▶ Redistribution $C_t^H = w_t N_t + T_t^H$

$$\lambda T_{H,t} = \tau^D D_t + \tau^K r_t^K K_t$$

- ▶ **No Y ineq.** $\chi = 1$: perfect redistribution

$$\tau^D = \tau^K = \lambda$$

Parameterization: Standard

Parameter	Value	Description
α	0.33	Capital share of output
δ	0.025	Depreciation rate per quarter
ω	10	Elasticity of investment to Q
β	0.99	Discount factor
s	1 / 0.98	Probability of staying unconstrained
σ	1	Intertemporal elasticity of substitution
$1/\varphi$	1.00	Frisch elasticity
λ	0 / 0.27	Share of hand-to-mouth
τ^D, τ^K	$= \begin{cases} 0 & \text{no redistribution} \\ \lambda & \text{full redistribution} \end{cases}$	Taxes on profits and capital
ψ	0.050	Slope of PC
ψ_w	∞ / 0.075	Slope of PC wages
ϕ_π	1.50	Taylor rule coefficient
ρ_i	0.60	Persistence MP shock

Quantifying the Complementarity

Amplification of Mon. Pol. Effects on C

Rep. Agent		Heterogeneous agents		
		No Y inequality	Y inequality	+ Risk
no- K	1	1(=RANK)	1.51	
K	0.66	1.11	2.25	

- RANK: K dampens, real (long) rate increases I response, less C

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- ▶ **Capital inequality** amplifies even under no income inequality

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- ▶ RANK: K dampens, real (long) rate increases I response, less C
- ▶ **Capital inequality** amplifies even under no income inequality
- ▶ **Income inequality** *can* lead to substantial amplification
- ▶ Channels are strongly **complementary**: $2.25 \gg 1.11 * 1.51$

Quantifying the Complementarity

Amplification of Mon. Pol. Effects on C

Rep. Agent		Heterogeneous agents		
		No Y inequality	Y inequality	+ Risk
no- K	1	1(=RANK)	1.51	1.60
K	0.66	1.11	2.25	2.62

- ▶ RANK: K dampens, real (long) rate increases I response, less C
- ▶ **Capital inequality** amplifies even under no income inequality
- ▶ **Income inequality** *can* lead to substantial amplification
- ▶ Channels are strongly **complementary**: $2.25 \gg 1.15 * 1.51$
- ▶ Idiosyncratic risk **reinforces complementarity** (prec. saving in liquid assets)

Analytical Solution

- Novel (incl. RANK!) $\delta = 1; \varphi = 0$; fix r_t (relax: paper):

$$\frac{\partial c_t}{\partial(-r_t)} = \frac{1-\lambda}{1-\lambda\chi_K} \left[1 + \frac{\lambda\alpha\beta}{(1-\lambda)(1-\alpha\beta)} \frac{(1-\alpha)}{1 + \alpha \frac{\lambda(\chi_K-1)}{1-\lambda\chi_K} \left(1 + \frac{1-\alpha\beta}{1-\alpha}\right)} \right]$$

$$\chi_K = 1 + \frac{1-\alpha}{1-\alpha\beta}$$

- complementarity proof

- K adjustment costs:

$$\frac{\partial k_{t+1}}{\partial(-r_t)} = \omega\mu_\omega; \quad \frac{\partial c_t}{\partial(-r_t)} = \frac{1-\lambda}{1-\lambda\chi_K} \left(1 + \omega\mu_\omega \frac{\lambda}{1-\lambda} \frac{\alpha\beta}{1-\alpha\beta} \right).$$

μ_ω stable root of 2nd order diff eq.

- big difference for K *Investment*, not for C *Consumption*!

- Consistent with Alves Kaplan Moll Violante HANK result.

Fiscal Redistribution

Income Redistribution		Profit Income D	
		Yes	No
Capital Income $r^K K$	Yes	1.11	4.34
	No	0.50	2.62

- ▶ (*Model*) Profits are countercyclical $\rightarrow Y_H$ *less* cyclical $\rightarrow$ dampening
- ▶ $r^K K$ strongly procyclical $\rightarrow Y_H$ *more* cyclical $\rightarrow$ ++*amplification*
- ▶ Less stark with **sticky wages**, profits less countercyclical

Sticky Wages

Amplification of Mon. Pol. Effects on C

	Rep. Agent	Heterogeneous Agents		
		No Y ineq.	Y ineq.	+ Risk
no- K	1.37 (37%)	1.37 (37%)	1.38 (-9%)	1.40 (-13%)
K	1.29 (95%)	2.10 (89%)	2.43 (8%)	2.67 (2%)

- **RA:** Further break of neutrality $\rightarrow$ more MP power

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K	1.29 (95%)	2.10 (89%)	2.43 (8%)	2.67 (2%)

- ▶ RA: Further break of neutrality → more MP power
- ▶ No Y ineq.: also **amplification**, but ...
 - ▶ conditional-on-K similar to flex-W ($\frac{2.1}{1.29} \sim \frac{1.11}{0.66}$)

Sticky Wages

Amplification of Mon. Pol. Effects on C

	Rep. Agent	Heterogeneous Agents		
		No Y ineq.	Y ineq.	+ Risk
no- K	1.37 (37%)	1.37 (37%)	1.38 (-9%)	1.40 (-13%)
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- ▶ RA: Further break of neutrality $\rightarrow$ more MP power
- ▶ No Y ineq.: also amplification (*but ...*)
- ▶ **Dampens** Y ineq. channel (w/o K !)

Sticky Wages

Amplification of Mon. Pol. Effects on C

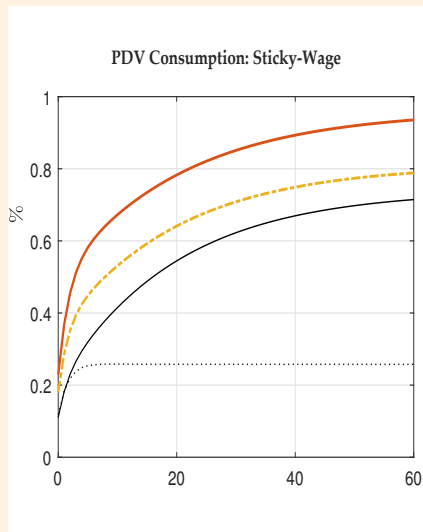
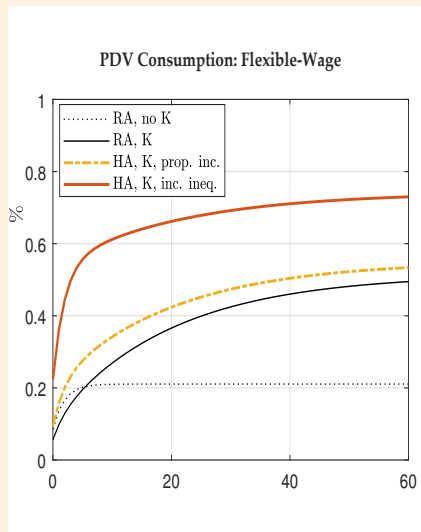
	RANK	No Y ineq.	Y ineq.	+ Risk
no- K	1.37 (37%)	1.37 (37%)	1.38 (-9%)	1.40 (-13%)
K	1.29 (95%)	2.10 (89%)	2.43 (8%)	2.67 (2%)

- ▶ RA: Further break of neutrality $\rightarrow$ more MP power
- ▶ No Y ineq.: also amplification (*but ...*)
- ▶ **Dampens** Y ineq. channel (w/o K !)
- ▶ **Complementarity** is robust
 - $\rightarrow$ w/ K : sticky wages key amplification mechanism only w/ proportional Y , not w/ *countercyclical Y inequality*

Conclusions

- ▶ Further step toward Macro convergence, 2 dimensions
 - ▶ RBC & NK
 - ▶ quantitative & tractable HANK
- ▶ K of ... capital importance for policy-relevant, monetary models
 - ▶ through *a multiplier of the multiplier*:
 - ▶ complementarity $K - Y$ inequality
- ▶ Key for MP what capital income is redistributed, physical vs profits
 - ▶ **optimal policy?**

Long-run Amplification



Table

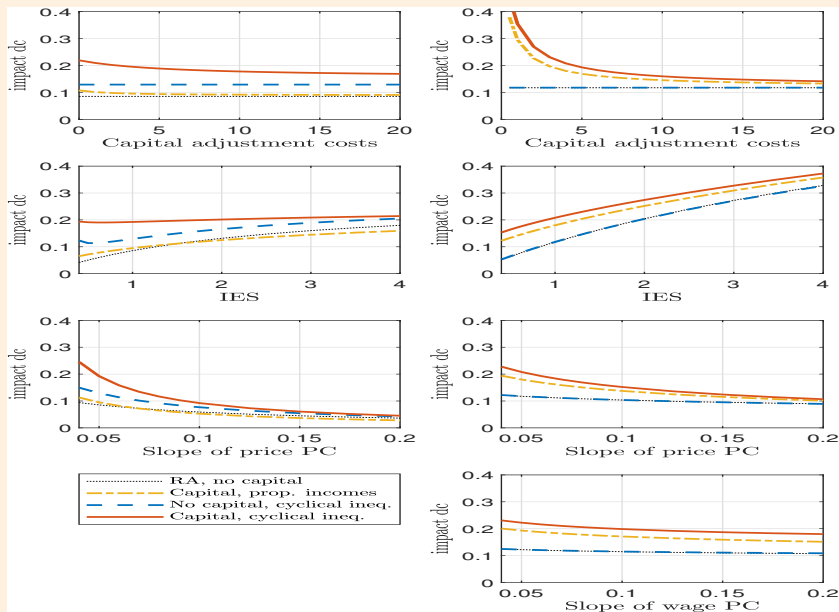


Figure:

Testable Prediction 1: Peak Responses to MP

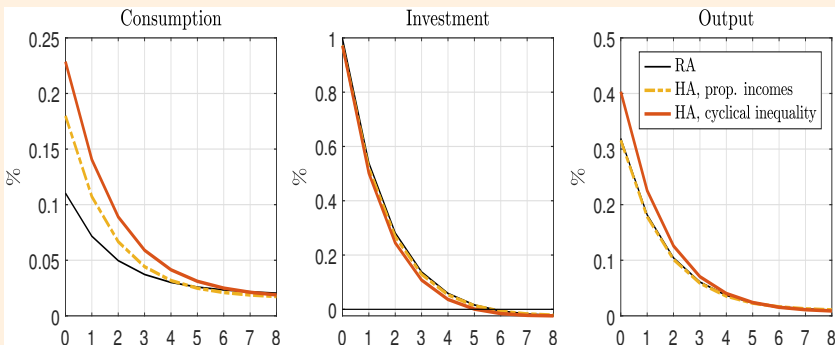


Figure: IRFs to expansionary interest rate shock of 25 basis points.

Testable Prediction(s) 2: C & Y Inequality

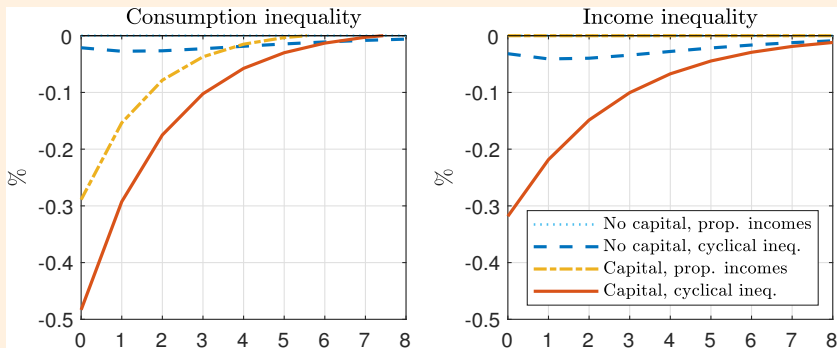


Figure: IRFs to expansionary interest rate shock of 25 basis points.

- **Only** HA with *both* K and Y inequality fits evidence by i.a. Coibion et al (2017)

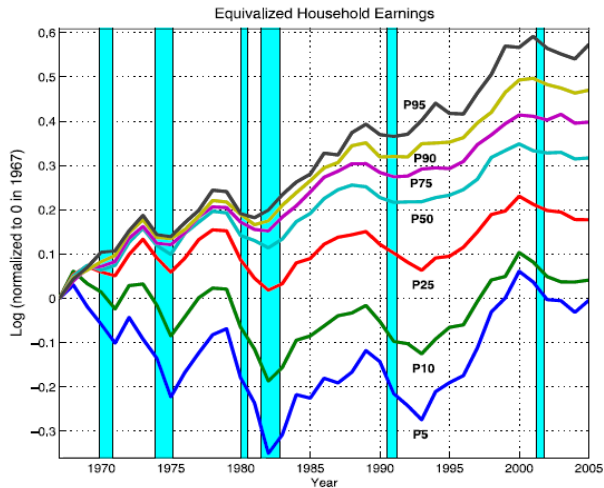
Empirical Evidence for Cyclicalities of Inequality?

Micro

- ▶ **Bottomline:** seems countercyclical $\chi > 1$
- ▶ Heathcote Storesletten Violante 2010 RED in Y distribution
- ▶ Guvenen et al 2004 JPE, etc: U-shaped, skewness, worker betas, etc.– income **risk** (Storesletten Telmer Yaron 2004, Mankiw 1986)
- ▶ Cloyne Ferreira Surico 2018 REStud: larger effect for mortgagors, through income
- ▶ Lenza Slacalek 2018 – response to MP by Y quantile (key: unemployment)
- ▶ Patterson 2019 JMP: first **direct** test, matching MPCs and individual Y cyclicalities
- ▶ Alves Kaplan Moll Violante 2019 (fct of permanent Y , Mincerian regressions)
- ▶ Slacalek Tristani Violante 2019: constrained (low **liquid** wealth) vs unconstrained

Heathcote, Perri, Violante RED 2010

J. Heathcote et al. / Review of Economic Dynamics 13 (2010) 15–51



Guvenen Ozkan Song 2014 JPE

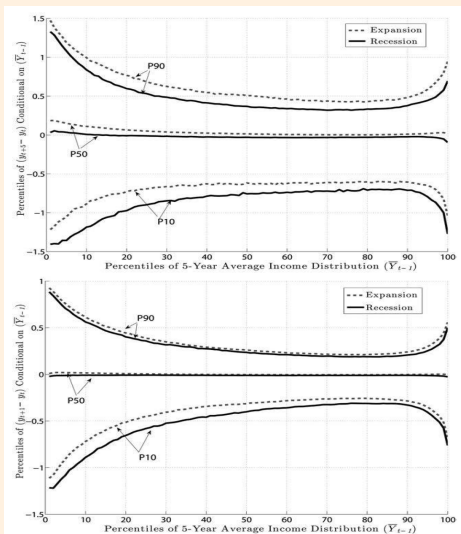
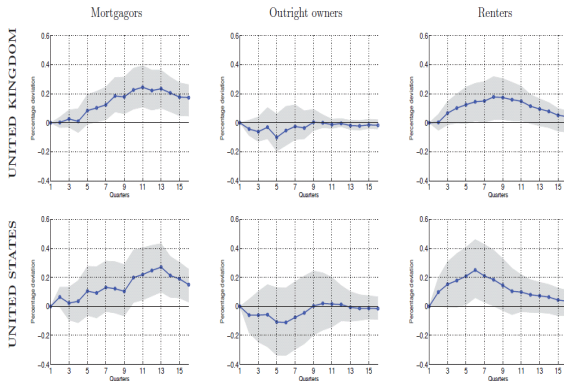


FIG. 7.—Percentiles of the earnings growth distribution: recession versus expansion. *Top*, persistent change. *Bottom*, transitory change.

1. What drives the heterogeneity in consumption?



*response
to a 25 bps
interest
rate cut*

Source: “Monetary Policy When Households have Debt” (Cloyne, Ferreira, Surico, 2018, ReStud).

5

2A. The indirect effects of MP through income

Table 1: CUMULATIVE CHANGES OVER FOUR YEARS IN US\$

Panel A: United Kingdom				
	<i>Non-durable consumption</i>	<i>Durable expenditure</i>	<i>Mortgage or rental payments</i>	<i>After-tax income</i>
Mortgagors	308.3 [112.8, 516.1]	292.3 [189.2, 369.0]	-166.4 [-272.2, -41.7]	695.9 [186.5, 1105.1]
Outright owners	-62.6 [-148.2, 77.4]	46.5 [-24.6, 107.6]		451.7 [122.5, 797.2]
Renters	155.3 [17.9, 261.8]	19.0 [-36.5, 62.9]	64.7 [4.4, 118.7]	397.3 [94.2, 596.1]

The unequal incidence channel of Bilbiie (2008, 2019): **AMPLIFICATION**

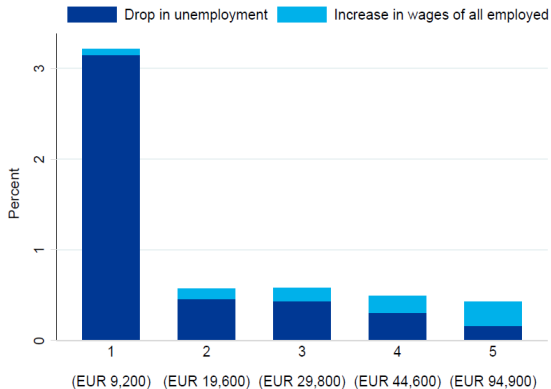
Source: "Monetary Policy When Households have Debt" (Cloyne, Ferreira, Surico, 2018, ReStud).

9

Source: P. Surico slides

Lenza Slacalek 2018

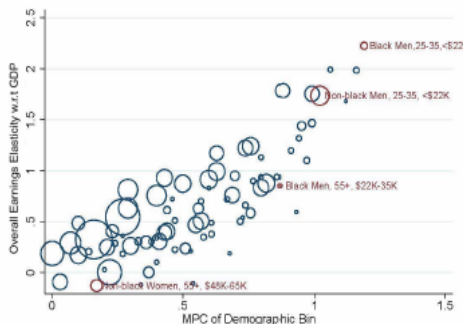
Response of income to MP easing (≈ 100 bp), by income quintile



Patterson 2019

First to do MPCs and " χ s"

Figure 1: Recession Exposure and MPC by Demographic Group



Notes: Sample includes the set of all workers employed in a sample state in year $t - 1$ from 1995 to 2011. The dependent variable in the regression producing the y-axis estimates is the total change in log earnings for the demographic group. The size of each bubble represents the earnings share of that demographic group. The coefficient on the fitted line for this plot is 1.33. Appendix Figure A14 shows the corresponding figure separately for the intensive and extensive margin of earnings.

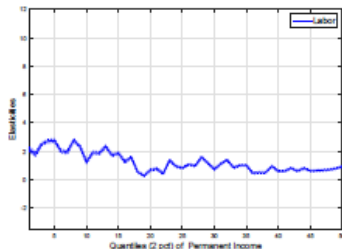
Slacalek Tristani Violante 2020

Table 2: Consumption–Income Ratios, Aggregate Responses to Monetary Policy Shock and Unequal Income Incidence

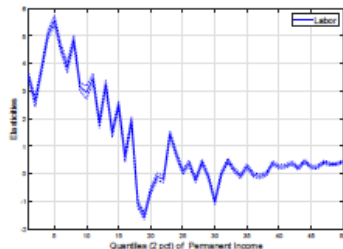
Variable	Country			
	Germany	Spain	France	Italy
Sensitivity of Household Employment to Aggregate Employment by HtM Status				
Poor Hand-to-Mouth	1.7	2.7	1.6	2.1
Wealthy Hand-to-Mouth	0.3	1.6	1.3	1.6
Non Hand-to-Mouth	1.1	0.8	0.9	0.8

See Fig. 15

Alves Kaplan Moll Violante 2019



(a) Elasticity using logarithm



(b) Elasticity using inverse hyperbolic sine

Figure 2: Estimated elasticities of individual earnings to aggregate earnings as a function of permanent income quantile. Dotted lines are the 95% confidence bands. Source: ASEC 1967-2017.

Optimal Monetary (Fiscal?) Policy in THANK

Optimal Policy in THANK

- ▶ First-best: perfect insurance. Trivial.
- ▶ Ramsey problem

$$\max_{\{C_t^H, C_t^S, N_t^H, N_t^S, \pi_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \{ \lambda U(C_t^H, N_t^H) + (1 - \lambda) U(C_t^S, N_t^S) \\ + \zeta_{j,t} \Xi_{j,t}$$

$\Xi_{j,t}$: Ramsey constraints (private equilibrium conditions)

$\zeta_{j,t}$ co-state Lagrange multipliers (with arbitrary initial values:
time-0 vs timeless).

▶

$$\begin{aligned} \Xi_{4,t} = & \pi_t(1 + \pi_t) - \beta E_t \left[\frac{U_C(\bar{C}_{t+1}^S)}{U_C(\bar{C}_t^S)} \frac{\lambda N_{t+1}^H + (1-\lambda)N_{t+1}^S}{\lambda N_t^H + (1-\lambda)N_t^S} \pi_{t+1}(1 + \pi_{t+1}) \right] \\ & + \frac{\varepsilon-1}{\psi} \left[\frac{\varepsilon}{\varepsilon-1} \frac{U_N(N_t^H)}{U_C(\bar{C}_t^H)} + 1 + \tau^S \right] \end{aligned}$$

- ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡ | ≡ ↺ 🔍 ↻

Optimal Policy in THANK: Ramsey Constraints



$$\mathbb{E}_{1,t} = \frac{U_N(N_t^S)}{U_C(C_t^S)} - \frac{U_N(N_t^H)}{U_C(C_t^H)}$$

$$\Xi_{2,t} = C_t^H + \frac{U_N(N_t^H)}{U_C(C_t^H)} N_t^H - \frac{\tau^D}{\lambda} (1 - \frac{\psi}{2} \pi_t^2 + \frac{U_N(N_t^H)}{U_C(C_t^H)}) (\lambda N_t^H + (1 - \lambda) N_t^S)$$

$$\mathbb{E}_{3,t} = \lambda C_t^H + (1 - \lambda) C_t^S - (1 - \frac{\psi}{2} \pi_t^2) (\lambda N_t^H + (1 - \lambda) N_t^S)$$

$$\begin{aligned} \mathbb{E}_{4,t} = & \pi_t(1 + \pi_t) - \beta E_t \left[\frac{U_C(\bar{C}_{t+1}^S)}{U_C(\bar{C}_t^S)} \frac{\lambda N_{t+1}^H + (1-\lambda)N_{t+1}^S}{\lambda N_t^H + (1-\lambda)N_t^S} \pi_{t+1}(1 + \pi_{t+1}) \right] \\ & + \frac{\varepsilon-1}{\psi} \left[\frac{\varepsilon}{\varepsilon-1} \frac{U_N(N_t^H)}{U_C(\bar{C}_t^H)} + 1 + \tau^S \right] \end{aligned}$$

- **Important:** Self-insurance NOT a constraint! ($\sim$ RANK)

$$U_C(C_t^S) = \beta E_t \left[\frac{1+i_t}{1+\pi_{t+1}} \left(s(C_{t+1}) U_C(C_{t+1}^S) + (1-s(C_{t+1})) U_C(C_{t+1}^H) \right) \right]$$

determines i_t residually once we found the allocation

- no longer true in other contexts, $i \rightarrow \text{MPC} \rightarrow \text{pass-through Y-C}$

Acharya Challe Dogra, in progress

Optimal Policy in THANK

- Optimal long-run inflation rate:

$$\pi^* = 0$$

- ▶ "easy" to show as SS of Ramsey problem (like RANK)
- ▶ Approx aggreg. welfare around **first-best, perfect-insurance** y^*

(Woodford 2003 RANK, Bilbiie 2008 **TANK**)⁴

$$\min_{\{c_t, \pi_t\}} \frac{1}{2} E_0 \sum_{t=0}^{\infty} \beta^t \left\{ \underbrace{\pi_t^2 + \alpha_y y_t^2}_{\text{RANK}} + \underbrace{\alpha_\gamma \gamma_t^2}_{\text{ineq.-THANK}} \right\},$$

$$\alpha_y \equiv (\sigma^{-1} + \varphi) / \psi; \quad \alpha_\gamma \equiv \lambda (1 - \lambda) \sigma^{-1} \varphi^{-1} \alpha_y$$

- note: more general, around target efficient y^* , change constraint – cost-push shocks

$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t + u_t,$$

⁴Relevant TANK extensions: Ascari et al sticky wages; Areosa Areosa different labor types; Nistico switching, wealth, financial regulation, etc. »  »  »  »  »

Optimal Policy in THANK

- ▶ key features: 1. no linear term; 2. recall γ prop. to y

$$\gamma_t = y_t^S - y_t^H = \frac{1 - \chi}{1 - \lambda} y_t$$

- ▶ **result:** risk *irrelevant* (around perf-insurance equil.)
- ▶ heterogeneity $\longrightarrow$ *less π stabilization* (key: **profits**) $\longrightarrow$ *more π volatility* under optimal policy

discretion:
$$\pi_t = -\frac{\alpha_y}{\kappa} \left(1 + \frac{\lambda}{1 - \lambda} \sigma^{-1} \varphi^{-1} (\chi - 1)^2 \right) y_t$$

- ▶ cyclicity of Γ irrelevant (note *square*) jump
- ▶ survives in quant-HANK: Bhandari Evans Golosov Sargent
- ▶ commitment: similar, but price-level targeting eventually
 - ▶ side remark: determinacy
- ▶ proportionality γ - y gap breaks down generally:
 - ▶ quantitative: Bhandari et al, Bilbiie Ragot, Legrand et al
 - ▶ analytical: Bilbiie Monacelli Perotti, Acharya Challe Dogra

The Analytics of Liquidity Traps with Heterogeneity

Warm-up: RANK Liquidity Traps

- ▶ Crash-course, simplest RANK: $\pi_t = \kappa c_t$ and $c_t = E_t c_{t+1} - \sigma (i_t - E_t \pi_{t+1} - \rho_t)$
- ▶ **Sunspot LTs** (Benhabib, Schmitt-Grohe, Uribe; Mertens Ravn ... Bilbiie)

$$c_L = \frac{\sigma}{1 - z_s \nu_0} \rho < 0 \text{ iff } z_s > \nu_0^{-1}$$

- ▶ **Fundamental LT** (Eggertsson-Woodford), $\rho_t = \rho_L < 0$ with z or absorbed $\rho_t = \rho > 0$:

$$c_L = \frac{\sigma}{1 - z \nu_0} \rho_L; \pi_L = \kappa c_L$$

- ▶ **Asymptote-bifurcation** "unbounded" recession $z \nu_0 \rightarrow 1$
- ▶ Deep recession and "**missing deflation**" (Hall)
- ▶ **Paradox of flexibility** (Eggertsson Krugman)

$$\partial \left(\frac{\partial c_L}{\partial \rho_L} \right) / \partial \kappa > 0$$

Liquidity Trap in THANK

- ▶ 3-equation simplest model, natural rate shock

$$c_t = \delta E_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} (i_t - E_t \pi_{t+1} - \rho_t)$$

$$i_t = \max(\rho_t, 0)$$

$$\pi_t = \kappa c_t$$

- ▶ can be generalized (Taylor, forward-looking PC, microfounded ρ shock)

Confidence-driven LT & neo-Fisherian equilibria

- Benhabib Schmitt-Grohe Uribe; Mertens and Ravn

$$c_L = \frac{\sigma^{\frac{1-\lambda}{1-\lambda\chi}}}{1 - z_s \nu_0} \rho; \quad \pi_L = \kappa c_L.$$

same

$$\nu_0 = \delta + \kappa \sigma \frac{1 - \lambda}{1 - \lambda \chi}$$

- Condition for sunspot LTs:

$$z_s > \nu_0^{-1}$$

→ impossible iff $\nu_0 < 1$ i.e.:

$$\chi < 1.$$

- BUT more likely with $\chi > 1$

Fundamental LT

- ▶ Eggertsson-Woodford: Markov chain $\rho_t = \rho_L < 0$ with z , absorbed $\rho_t = \rho > 0$ with $1 - z$.
- ▶ Equilibrium ($\pi_L = \kappa c_L$):

$$c_L = \frac{\sigma \frac{1-\lambda}{1-\lambda\chi}}{1 - z\nu_0} \rho_L;$$

- ▶ No asymptote/bifurcation (bounded recessions and multipliers) if

$$\nu_0 < 1.$$

- ▶ Deep recessions, multipliers, ZLB paradoxes?

Amplification: Deeper (and Deflationless) Recessions?

- Equilibrium ($\pi_L = \kappa c_L$):

$$c_L = \frac{\sigma \frac{1-\lambda}{1-\lambda\chi}}{1 - z\nu_0} \rho_L;$$

- Amplification/dampening channels:

$$\sigma \frac{1-\lambda}{1-\lambda\chi}, \delta, \kappa \sigma \frac{1-\lambda}{1-\lambda\chi}$$

- Even as $\kappa \rightarrow 0$ (!), $\sigma \frac{1-\lambda}{1-\lambda\chi}, \delta$ channels still work...iff:

$$\text{CI: } \chi > 1$$

- Transcends simple model, see very rich globally-solved
HANK+ZLB 2021 Fernandez-Villaverde Nuno Rachedi

Amplification: Inflationless Fiscal Multipliers?

- ▶ **HANK multiplier** (balanced-budget uniform $t_t = g_t$):⁵

$$\mathcal{M}_c \equiv \frac{\zeta}{1 - v_0 z} \left[\underbrace{(\chi - 1) \frac{(1 - z) \lambda + z (1 - s)}{1 - \lambda \chi}}_{\text{TANK + HANK AD}} + \underbrace{\kappa \sigma \frac{1 - \lambda}{1 - \lambda \chi} z}_{\text{RANK AS, E}(\pi)} \right].$$

- ▶ separate: progressivity of the essence (transfer multiplier positive)
- ▶ Amplification condition:

$$\chi > 1$$

⁵Income effect $\zeta \equiv (1 + \varphi^{-1} \sigma^{-1})^{-1}$

Paradox of Flexibility: Eggertsson Krugman

- (Tobin 75, De Long Summers 86): $\kappa \uparrow$ makes ZLB recession worse $c_L \downarrow$

$$c_L = \frac{\sigma \frac{1-\lambda}{1-\lambda\chi}}{1 - z\nu_0} \rho_L; \pi_L = \kappa c_L$$

$$\partial \left(\frac{\partial c_L}{\partial \rho_L} \right) / \partial \kappa = z \left(\frac{\sigma \frac{1-\lambda}{1-\lambda\chi}}{1 - z\nu_0} \right)^2$$

- Mitigate paradox if decreasing with λ , i.e. if $\sigma \frac{1-\lambda}{1-\lambda\chi}$ and δ decreasing with λ :

$$\chi < 1.$$

Forward Guidance in a Liquidity Trap in THANK

- ▶ use Bilbiie 2016 (*Optimal Forward Guidance*, AEJ-Macro)
simple closed-form, FG "state"
 - ▶ **first closed-form optimal policy** in RANK-LT ($\sim$ Ramsey) +
"simple rule FG" (how long should CB $i = 0$)
- ▶ Extend Eggertsson-Woodford to *three states*:
 $P(L \rightarrow F) = (1 - z)q$, $P(F \rightarrow F) = q$; $E(\text{FG duration}) = 1 / (1 - q)$

$$E_t c_{t+1} = z c_L + (1 - z) q c_F + (1 - z) (1 - q) 0$$

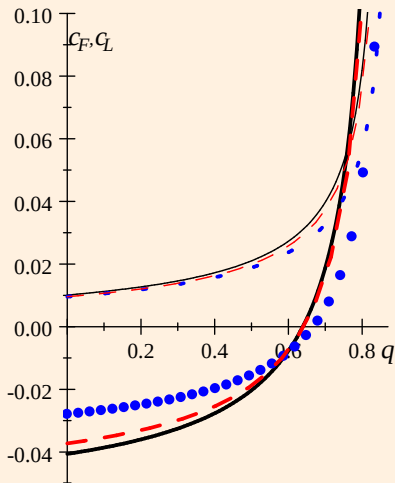
- ▶ Equilibrium $dc_F/dq > 0$; $dc_L/dq > 0$

$$c_F = \frac{1}{1 - q\nu_0} \sigma \frac{1 - \lambda}{1 - \lambda\chi} \rho$$

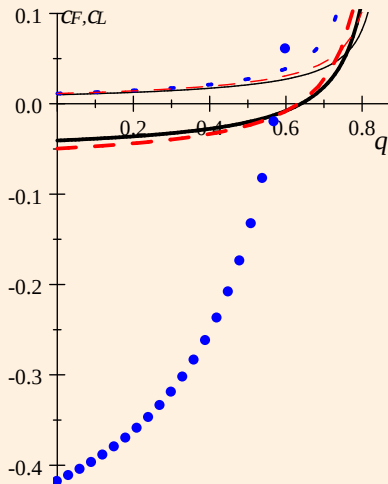
$$c_L = \frac{1 - z}{1 - z\nu_0} \frac{q\nu_0}{1 - q\nu_0} \sigma \frac{1 - \lambda}{1 - \lambda\chi} \rho + \frac{\sigma}{1 - z\nu_0} \frac{1 - \lambda}{1 - \lambda\chi} \rho_L$$

FG: Dampening and Amplification

c_L (thick) and c_F (thin): RANK, TANK and iid-HANK



PI: $\chi < 1$



CI: $\chi > 1$

FG Power and Puzzle

► **FG power:**

$$\mathcal{P}_{FG} \equiv \frac{dc_L}{dq} = \left(\frac{1}{1 - q\nu_0} \right)^2 \frac{(1 - z) \nu_0 \sigma^{\frac{1-\lambda}{1-\lambda\chi}}}{1 - z\nu_0} \rho.$$

► $\chi > 1: \partial \mathcal{P}_{FG} / \partial \lambda > 0; \partial \mathcal{P}_{FG} / \partial (1 - s) > 0$

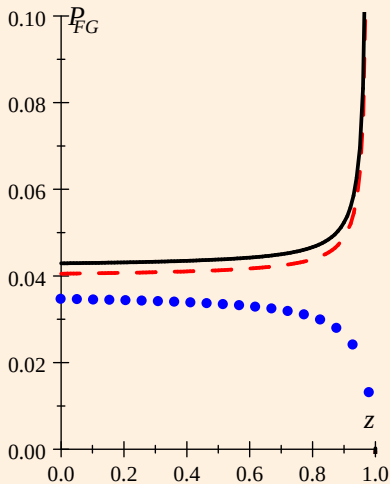
► *Corollary:*

FG puzzle: $\frac{\partial \mathcal{P}_{FG}}{\partial z} \geq 0$ ruled out *iff*

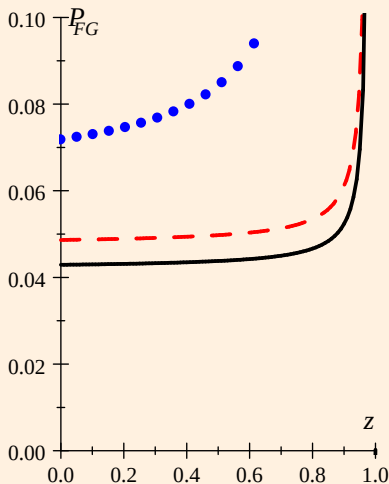
$$\nu_0 < 1.$$

FG Puzzle: Resolution or Aggravation?

RANK, TANK and iid-HANK; $q = 0.5$ $\lambda = 0.1$



PI: $\chi < 1$



CI: $\chi > 1$

Optimal Policy in LT (and the Dark Side of FG Power)

- ▶ **RANK:** Eggertsson Woodford 2003, Jung Teranishi Watanabe 2005, Nakov 2008, Adam Billi 2008, Nakata, Schmidt, Nakata Schmidt, Bilbiie 2016 (analytical)

- ▶ $E(PDV(\text{Welfare}))$ w / Markov chain:

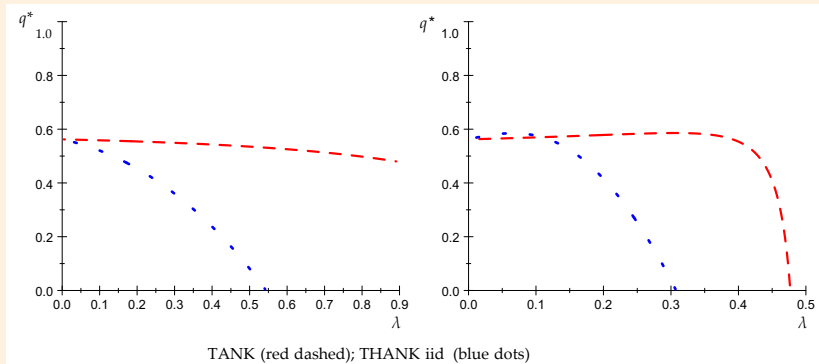
$$W = \frac{1}{1 - \beta z} \frac{1}{2} \left[c_L^2 + \omega(q) c_F^2 \right],$$

- ▶ $\omega(q) = \frac{1 - \beta z + \beta(1 - z)q}{1 - \beta q}$, $\omega'(q) > 0$: the longer in F, the larger the total welfare cost.
- ▶ $\min_q W$ s.t. equilibrium c_F and c_L

$$c_L \frac{dc_L}{dq} + \omega(q) c_F \frac{dc_F}{dq} + \frac{1}{2} \frac{d\omega(q)}{dq} c_F^2 = 0$$

- ▶ simple case: closed-form q^* (paper)

Optimal FG duration



$q^*(\lambda)$: $\chi < 1$ (left) and $\chi > 1$ (right)

Further Developments and Extensions

1. Money - Liquidity and Optimal Mon. Pol. (w/ Ragot)
2. Stabilization v Redistribution: Optimal Mon-Fiscal Mix (w/ Monacelli & Perotti)
3. Capital - (Illiquid) Wealth Inequality (w/ Känzig and Surico)
4. Macro Estimation (w/ Primiceri and Tambalotti)

Money - Liquidity

Based on Bilbiie and Ragot:

*Optimal Monetary Policy and Liquidity with
Heterogeneous Households*

Money:

- ▶ THANK with money as only liquid asset
- ▶ literally Bewley model (with extreme Baumol-Tobin flavor)
- ▶ Self-insurance

$$c_t^S = s\beta E_t c_{t+1}^S + (1 - s\beta) E_t c_{t+1}^H + \sigma E_t \pi_{t+1}$$

- ▶ money injected through helicopter drops (paper: open-market, swap with illiquid assets)
- ▶ Loglinearized equil., m is SS money demand M/P as share of C (inv. velocity); SS needs to be unequal/distorted $C^H < C^S$ for there to be a well-defined money demand

Money:

- Imperfect *Insurance (inequality)* γ_t and *Liquidity* x :

$$\gamma_t^c \equiv c_t^S - c_t^H = -\frac{s - (1 - \lambda)}{\lambda} x_t$$

$s - (1 - \lambda)$: precautionary saving through *liquidity*

Optimal MP w/ Liquidity

- ▶ liquidity-insurance motive
- ▶ linear term in loss function, no longer LQ problem
- ▶ complementarity with NK monopoly distortion
- ▶ large deviations from price stability

Optimal Policy: First-best and Flex prices

► Reiterate *known flex-price results*:

1. Perfect insurance *first-best*, recover **Friedman rule**(-ish)
2. **But** inflation ("Pigou effect") for *insurance*

$$C_t^H = Y_t^H + X_t + \frac{1-s}{\lambda} \frac{M_t}{1+\pi_t}$$

3. What when costly/distortionary (sticky prices)
 - tolerate some (insuring) inflation even when costly.

Optimal Policy: Ramsey problem (HD)

- ▶ Same as before otherwise
- ▶ Money and nominal interest orthogonal—only HD money issuance; more complicated with open-market (paper)

Optimal Policy: Insurance vs Stabilization

$$\min \mathcal{L} = \underbrace{\pi_t^2 + \alpha_y c_t^2}_{\text{RANK}} + \underbrace{\alpha_\gamma \gamma_t^2}_{\text{ineq.}} - \underbrace{2 \frac{\lambda \bar{c}^H (\Gamma^{\frac{1}{\sigma}} - 1)}{\psi} \hat{c}_t^H}_{\text{Novel}}$$

- ▶ **Novel** Liquidity-*Insurance* motive $\Gamma = \frac{1+\pi-s}{\beta} = 1 + \frac{i}{1-s} > 1$
- ▶ **first-order** gap matters 2nd order when Γ large "enough"
 - ▶ ~inefficient SS and monopolistic distortion in NK, Benigno Woodford
 - ▶ vanishes at Friedman rule, $\Gamma \rightarrow 1$ (also for $\beta \rightarrow 1$, etc.)
- ▶ previously: SS insurance $\Gamma = 1$ Bilbiie 08, 18; Curdia Woodford 16; Nistico 16
- ▶ Bottomline: **not** LQ problem, solve **full Ramsey**

Optimal Deviations from Price Stability: Long Run

- ▶ **Long-run** standard monetary NK optimal π^*

$$\text{Friedman rule} \leq \pi^* \leq 0$$

- ▶ **Calibration 1:** match data

$$(\lambda = 0.5; m = 0.5; \Delta c^{H-S} = 10\%) \rightarrow \\ (s = 0.9; h = 0.9; \delta_c = 0.78)$$

- ▶ robust to labor market ("Shimer") calibration
($s = 0.95; h = 0.5; \delta_c = 0.5$)

- ▶ $\pi^* = -.79\%$

- ▶ Welfare costs: moving from annual $\pi = 2\%$ to $\pi^* \rightarrow$
 $\Delta W(c) = .61\%$ (~Lucas, Imrohoroglu)

Optimal Deviations from Price Stability: Short Run

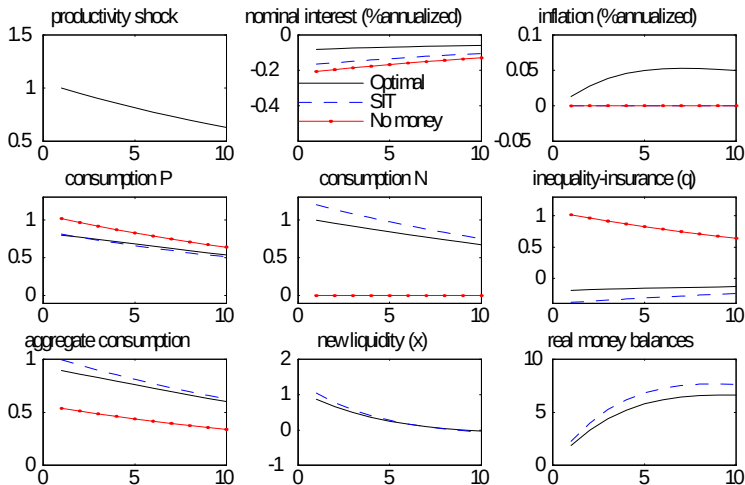


Figure 1: Responses to a labor productivity: Ramsey (solid black), SIT (blue dash), and Ramsey in cashless model (red circles).

- ▶ monetary policy is "procyclical"! — to provide insurance (boom generates lack of insurance)
- ▶ more liquidity, more insurance (interest rate goes down).
- ▶ redistribution dynamics: inflation "bad" for H , but m dominates
- ▶ use inflation optimally

Optimal Deviations from Price Stability: Short Run

Economies	Std dev (%)				Welfare (%)
	c^S	c^H	γ	π	Δ^W
Baseline					
Ramsey	2.6	3.3	.6	.05	—
SIT	2.6	3.7	1.2	0	.08
NK Cross χ (endog N income)					
Ramsey	2.8	3.3	1.1	.02	—
SIT	2.8	3.5	1.4	0	.02
Ramsey certain					.001
Competitive: $\bar{L}$ & $\Delta m = 0$					.024
SS insurance ($\beta \rightarrow 1$)					
Ramsey	2.5	3.2	.6	.05	—
SIT	2.5	3.8	1.3	0	.00

reassuringly also in (richer) quant-HANK: Bhandari, Evans, Golosov, Sargent

Optimal Policy and Welfare Costs: Robustness

Economies	π^*		SD (%)		γ	π	Δ^W
			c^S	c^H			
More flex price	-1.54	Ramsey	2.7	3.2	.5	.08	—
$\psi = 50$		SIT	2.6	3.8	1.2	0	.13
Less elastic labor	-.6	Ramsey	2.1	2.7	.6	.04	—
$\varphi = 1$		SIT	2.0	3.0	.9	0	.03
Shimer ⁶	-.36	Ramsey	3.0	5.1	2.1	.02	—
		SIT	3.1	5.5	2.4	0	.06

Table 3: Robustness analysis

⁶Labor market ($\alpha = .95$; $\rho = .5$; $\delta = .5$)

Complementarity: Liquidity-Insurance – NK

Optimal Deviations from SIT: **Large NK (Monop.) Distortion**

Economies	$\pi^{SS}(\%)$	$sd(\pi)(\%)$	Δ_0^W
Baseline	-1.54	.06	.48
$\psi = 50$	-2.6	.1	.55
$\varphi = 1$	-0.1	.04	.15
$\beta \rightarrow 1$	-.7	.05	0
Shimer	-.6	.02	.33
NK Cross χ	-.4	.02	.16

Table 4: A distorted steady state, $\sigma = 0$

Ramsey-Optimal Policy with Open Market

- *paper*

Conclusions for Opt. Mon. Pol. w/ Liquidity

1. Novel *Liquidity-Insurance* motive
2. Optimal inflation volatility (large welfare costs of $\pi_t = 0$)
 - ▶ especially when NK distortions also large: **complementarity**
3. *How* to liquidity-insure? HD $\succsim$ OM

Stabilization vs Redistribution: The Optimal Monetary-Fiscal Mix*

Based on Homonymous paper with
Tommaso Monacelli and Roberto Perotti

*supersedes 2011(!) *Tax Cuts vs Government Spending: Welfare & the
ZLB*

This paper

- ▶ main point: **unequal incidence breaks divine coincidence**

1. M&F interlinked w/ HA: stabilization–redistribution
interaction non-trivial

2. Provide **examples**:

- ▶ benchmarks, not interact: equivalence, separation (optimal transfers)
- ▶ *a fortiori* interact (e.g. spending with unequal taxation incidence)

	Optimal MP	... and FP
Transfers	Exogenous	Endogenous
Gov.t Spending	Exogenous	Endogenous

Analytical TANK: Off the shelf

- ▶ Bilbiie 2008, 2020; Bilbiie Monacelli and Perotti 2013, etc.
- ▶ **Spell out fiscal side for completion**
- ▶ $U(C_t^j, N_t^j, G_t^j)$ separable
 - ▶ $\sigma^{-1} \equiv -U_{CC}^j C^j / U_C^j$; $\varphi \equiv U_{NN}^j N^j / U_N^j$; $\sigma_G^{-1} \equiv -U_{GG}^j G^j / U_G^j$
- ▶ absent frictions, "Samuelson rule" $U_C^j(C) = U_G^j(G)$.
 - ▶ Keep in *steady state*, study deviations w/ frictions

Analytical TANK: Off the shelf

- ▶ λ consume all **their** income $C_t^H = W_t N_t^H - T_t^H + \frac{\tau^D}{\lambda} \mathcal{D}_t$

Analytical TANK: Off the shelf

- ▶ λ consume all **their** income $C_t^H = W_t N_t^H - T_t^H + \frac{\tau^D}{\lambda} \mathcal{D}_t$
- ▶ S: (like complete markets)

$$C_t^S + P_t^{-1} B_t^S = P_t^{-1} B_{t-1}^S (1 + i_{t-1}) + W_t N_t^S - T_t^S + \frac{1 - \tau^D}{1 - \lambda} \mathcal{D}_t$$

Analytical TANK: Off the shelf

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- ▶ Labor supply: $-U_N^j(N_t^j) = W_t U_C^j(C_t^j), j = H, S.$

Analytical TANK: Off the shelf

- ▶ λ consume all **their** income $C_t^H = W_t N_t^H - T_t^H + \frac{\tau^D}{\lambda} \mathcal{D}_t$
- ▶ S: (like complete markets)

$$C_t^S + P_t^{-1} B_t^S = P_t^{-1} B_{t-1}^S (1 + i_{t-1}) + W_t N_t^S - T_t^S + \frac{1 - \tau^D}{1 - \lambda} \mathcal{D}_t$$

- ▶ Labor supply: $-U_N^j(N_t^j) = W_t U_C^j(C_t^j), j = H, S.$
- ▶ Euler: $U_C^S(C_t^S) = \beta \mathbb{E}_t \left[\frac{1+i_t}{\Pi_{t+1}} U_C^S(C_{t+1}^S) \right], \Pi_t$ gross inflation

Analytical TANK: Off the shelf

- ▶ λ consume all **their** income $C_t^H = W_t N_t^H - T_t^H + \frac{\tau^D}{\lambda} \mathcal{D}_t$
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$$C_t^S + P_t^{-1} B_t^S = P_t^{-1} B_{t-1}^S (1 + i_{t-1}) + W_t N_t^S - T_t^S + \frac{1 - \tau^D}{1 - \lambda} \mathcal{D}_t$$

- ▶ Labor supply: $-U_N^j(N_t^j) = W_t U_C^j(C_t^j), j = H, S.$
- ▶ Euler: $U_C^S(C_t^S) = \beta \mathbb{E}_t \left[\frac{1+i_t}{\Pi_{t+1}} U_C^S(C_{t+1}^S) \right], \Pi_t$ gross inflation
- ▶ Firms: $Y_t(i) = N_t(i), MC = W_t.$ Rotemberg $\rightarrow$ PC:

$$\Pi_t(\Pi_t - 1) = E_t(\Lambda_{t,t+1}^S \frac{Y_{t+1}}{Y_t} \Pi_{t+1}(\Pi_{t+1} - 1)) + \frac{\theta}{\psi} (W_t - \frac{(1 + \tau^S)(\theta - 1)}{\theta})$$

Analytical TANK: Off the shelf

- ▶ λ consume all **their** income $C_t^H = W_t N_t^H - T_t^H + \frac{\tau^D}{\lambda} \mathcal{D}_t$
- ▶ S: (like complete markets)

$$C_t^S + P_t^{-1} B_t^S = P_t^{-1} B_{t-1}^S (1 + i_{t-1}) + W_t N_t^S - T_t^S + \frac{1 - \tau^D}{1 - \lambda} \mathcal{D}_t$$

- ▶ Labor supply: $-U_N^j(N_t^j) = W_t U_C^j(C_t^j), j = H, S.$
- ▶ Euler: $U_C^S(C_t^S) = \beta \mathbb{E}_t \left[\frac{1+i_t}{\Pi_{t+1}} U_C^S(C_{t+1}^S) \right], \Pi_t$ gross inflation
- ▶ Firms: $Y_t(i) = N_t(i), MC = W_t.$ Rotemberg $\rightarrow$ PC:

$$\Pi_t(\Pi_t - 1) = E_t(\Lambda_{t,t+1}^S \frac{Y_{t+1}}{Y_t} \Pi_{t+1}(\Pi_{t+1} - 1)) + \frac{\theta}{\psi} (W_t - \frac{(1 + \tau^S)(\theta - 1)}{\theta})$$

- ▶ Aggregation:

$$Y_t \left[1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right] = \lambda C_t^H + (1 - \lambda) C_t^S + G_t.$$

Transfers Only

Pure transfer

- ▶ *government spending is zero:*

$$G_t = T_t = 0 \rightarrow \lambda T_{H,t} = -(1 - \lambda) T_{S,t}$$

- ▶ Let $t_{j,t} \approx (T_t^j - 0) / Y$:

$$t_{H,t} \equiv -f_t \text{ and } t_t^S = \frac{\lambda}{1 - \lambda} f_t$$

- ▶ **exogenous redistribution**, contrast with:
- ▶ **endogenous redistribution** $\tau^D / \lambda \sim$ progressivity of income tax system

Endo- & Exo-genous Redistribution: AD in TANK

- ▶ Loglin around $D = 0$: $c_t^H = w_t + n_t^H + \frac{\tau^D}{\lambda} d_t$. Equilibrium:

$$c_t^H = \chi c_t + \zeta f_t, \quad \chi \equiv 1 + \varphi \left(1 - \frac{\tau^D}{\lambda}\right); \quad \zeta \equiv \left[1 + (\varphi\sigma)^{-1}\right]^{-1}$$

$$c_t^S = \frac{1 - \lambda\chi}{1 - \lambda} c_t - \frac{\lambda}{1 - \lambda} \zeta f_t$$

Extra income effect $w \uparrow \rightarrow d \downarrow$ *keystone*: profits

$$\text{Cyclical Income Inequality: } \gamma_t \equiv c_t^S - c_t^H = \frac{1 - \chi}{1 - \lambda} c_t - \frac{\zeta}{1 - \lambda} f_t$$

- ▶ $\chi \sim$ income-distribution model: here, **example**
- ▶ Aggregate, replace $c_t^j \rightarrow c_t^S = E_t c_{t+1}^S - \sigma r_t$

A Monetary-Fiscal AD-Equivalence Proposition

- ▶ Echoes Bilbiie, Monacelli and Perotti (2013, eq. 24)

$$c_t = \mathbb{E}_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda \chi} (i_t - \mathbb{E}_t \pi_{t+1}) + \frac{\lambda \zeta}{1 - \lambda \chi} (f_t - \mathbb{E}_t f_{t+1})$$

- ▶ **Monetary-Fiscal AD Equivalence Proposition.**

$$r_t \equiv i_t - \mathbb{E}_t \pi_{t+1}$$

$$\sigma (1 - \lambda) (-r_t) = \lambda \zeta (f_t - \mathbb{E}_t f_{t+1}).$$

Direct, partial-equilibrium effects on AD

- ▶ GE: "NK cross" $\lambda \chi \simeq$ aggreg. MPC (Bilbiie 2020, Amplif. iff $\chi > 1$)
- ▶ Transcends this: richer HANK alas λ, χ , fiscal redistrib. (endo&exo).
 - ▶ Wolf (2021) general version, rich HANK
- ▶ Corollary: transfers undo ZLB, Correia et al in RANK; Egertsson Krugman

Inequality, Gaps, and Tradeoffs

- ▶ Phillips curve:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa c_t, \quad \kappa \equiv \theta \left(\sigma^{-1} + \varphi \right) / \psi.$$

- ▶ Flex-price equilibrium:

$$c_t^* = 0 \text{ and } \gamma_t^* = -\frac{\zeta}{1-\lambda} f_t$$

- ▶ **Key:** Transfers $\rightarrow$ inefficient γ_t even w/ flex prices (efficient SS $C^j = C!$)

- ▶ Perfect-insurance allocation:

$$\gamma_t^{**} = 0 \text{ and } c_t^{**} = -\frac{\zeta}{\chi-1} f_t$$

- ▶ Transfers = time-varying gap c_t^* vs c_t^{**} ($f_t = 0 \rightarrow$ "divine coincidence").

Inequality, Gaps, and Tradeoffs

- ▶ f_t exog. persist. p , perfect insurance $\rightarrow$ recession & deflation
 $w/\chi > 1$

$$\pi_t^{**} = -\frac{\kappa}{1-\beta p} \frac{\zeta}{\chi-1} f_t$$

- ▶ Why? Transfer $\rightarrow c \uparrow \rightarrow \gamma \downarrow\downarrow$. Engineer recession, $\gamma \uparrow 0$

$$(\text{close distributional gap}) \gamma_t = \frac{1-\chi}{1-\lambda} c_t - \frac{\zeta}{1-\lambda} f_t$$

1. Direct effect through f .
2. Indirect c response (depending on MP)
3. $\rightarrow$ feeds back into γ (depending on χ)

- ▶ Inherent tension: *supply-side* π vs *demand-side* γ distortions
- ▶ Essence, richer models w/ some akin inequality-AD feedback modulated by fiscal policy.

Optimal Policy w/ Exogenous Transfers

- ▶ Ramsey problem $\Leftrightarrow$ min quadratic loss $\mathcal{L}_t$:

$$\mathcal{L}_t \equiv \frac{1}{2} \left\{ \pi_t^2 + \underbrace{\frac{\sigma^{-1} + \varphi}{\psi} c_t^2}_{\text{output gap}} + \underbrace{\frac{(\sigma\zeta)^{-1}}{\psi} \lambda (1 - \lambda) \gamma_t^2}_{\text{inequality}} \right\}$$

s.t. $\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa c_t$ and

$$\gamma_t = \frac{1 - \chi}{1 - \lambda} c_t - \frac{\zeta}{1 - \lambda} f_t$$

- ▶ γ_t itself not novel (Bilbiie 2008, 2020 etc.)
- ▶ but $f_t =$ inefficiency wedge novel

Optimal Policy w/ Exogenous Transfers

$$\mathcal{L}_t \equiv \frac{1}{2} \left\{ \pi_t^2 + \frac{\sigma^{-1} + \varphi}{\psi} c_t^2 + \frac{(\sigma\zeta)^{-1}}{\psi} \frac{\lambda(\chi - 1)^2}{1 - \lambda} \left(c_t + \frac{\zeta}{\chi - 1} f_t \right)^2 \right\}$$

$$\text{s.t. } \pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa c_t$$

- Important: Euler residually determines i .

$$c_t = \mathbb{E}_t c_{t+1} - \sigma \frac{1 - \lambda}{1 - \lambda\chi} (i_t - \mathbb{E}_t \pi_{t+1}) + \frac{\lambda\zeta}{1 - \lambda\chi} (f_t - \mathbb{E}_t f_{t+1})$$

- Not in HANK: MPC fct of i , etc. – Acharya Challe Dogra 2020.
- **Discretion** throughout.
 - Timeless-optimal commitment immediate + *discretion hard* in quantitative models

Optimal Policy w/ Exogenous Transfers

- Solution:

$$c_t = -(\chi - 1) \frac{\lambda}{1 - \lambda} \frac{\Phi}{1 + \sigma\varphi} f_t; \quad \pi_t = \frac{\kappa}{1 - \beta p} c_t,$$

$$\gamma_t = -\frac{\zeta}{1 - \lambda} \left(1 + \frac{\kappa\theta}{1 - \beta p} \right) \Phi f_t, \quad \Phi > 0$$

- $\chi > 1$: $f_t > 0 \rightarrow$ contraction & deflation + inequality $\downarrow$.
- *Cet. par.* $\underline{c \uparrow}$ suboptimal $\rightarrow \gamma \downarrow$, compensate by $c \downarrow$, $\pi \downarrow \rightarrow \gamma \uparrow 0$
- Mitigate distributional gap:

$$c_t - c_t^{**} = \frac{\zeta}{\chi - 1} \left(1 + \frac{\kappa\theta}{1 - \beta p} \right) \Phi f_t.$$

- f_t optimally *deflationary*, expansionary rel. to c^{**} : gap > 0 (< 0 wrt c^*)
- *Negative comovement*: π & $c - c^{**}$ (NK jargon: "cost-pull")

Opt. Pol. w/ Endog. Transfers: Separation Proposition

- ▶ Exogenous ("cost-push") shocks u_t :

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa c_t + u_t$$

- ▶ Optimal f_t targeting rule:

$$c_t + \frac{\zeta}{\chi - 1} f_t = 0$$

- ▶ **Separation Proposition:** MP stabilizes $u, f_t \rightarrow$ perfect insurance
 - ▶ Surprise? Classic Tinbergen-Theil: instruments = targets.
- ▶ $\rightarrow$ RANK optimal-policy equilibrium! (Clarida et al 1999, Woodford 2003):

$$c_t = -\frac{\theta}{\theta\kappa + 1 - \beta p} u_t; \quad \pi_t = \frac{1}{\theta\kappa + 1 - \beta p} u_t$$
$$f_t = \frac{\chi - 1}{\zeta} \frac{\theta}{\theta\kappa + 1 - \beta p} u_t.$$

- ▶ Use redistribution f_t to stabilize γ : $\chi > 1, y \downarrow \rightarrow \gamma \uparrow \rightarrow f_t \uparrow$

Government Spending

Government Spending



$$G_t = T_t = \lambda T_t^H + (1 - \lambda) T_t^S$$

- ▶ Tax incidence, "progressivity":

$$\lambda T_t^H = \alpha T_t.$$

- ▶ Decomposition ($t_{H,t} \approx (T_t^H - G) / Y$, $g_t \approx (G_t - G) / Y$):

$$t_{H,t} = \frac{\alpha}{\lambda} t_t = \frac{\alpha}{\lambda} g_t = \underbrace{g_t}_{\text{uniform}} - \underbrace{\left(1 - \frac{\alpha}{\lambda}\right) t_t}_{\text{exogenous redistribution}}$$



lexible-price allocation, let

$$\tilde{g}_t \equiv \frac{1}{1-G_Y} g_t; \tilde{\varphi} \equiv \varphi (1 - G_Y); G_Y = G/Y$$

$$c_t^* = -\tilde{\zeta} \tilde{g}_t, \tilde{\zeta} \equiv \left[1 + (\tilde{\varphi} \sigma)^{-1}\right]^{-1}$$

Government Spending

- Bilbiie 2020, *NK Cross*, section 3.2:

$$c_t^H = \chi c_t - \tilde{\zeta} t_{H,t} = \chi c_t + \tilde{\zeta} \left(\chi - \frac{\alpha}{\lambda} \right) \tilde{g}_t$$

Aggregate Euler:

$$c_t = \mathbb{E}_t c_{t+1} + \frac{\lambda}{1 - \lambda \chi} \tilde{\zeta} \left(\chi - \frac{\alpha}{\lambda} \right) (\tilde{g}_t - \mathbb{E}_t \tilde{g}_{t+1}) - \sigma \frac{1 - \lambda}{1 - \lambda \chi} (i_t - \mathbb{E}_t \pi_{t+1})$$

- **Monetary-Fiscal AD Equivalence Proposition**

$$\sigma (1 - \lambda) (-r_t) = \lambda \tilde{\zeta} \left(\chi - \frac{\alpha}{\lambda} \right) (\tilde{g}_t - \mathbb{E}_t \tilde{g}_{t+1})$$

Same intuition (direct effects, etc.), undo ZLB, etc.

- Difference: g effects depend on α vs λ and χ :
 - exog. & endog. redistribution $\rightarrow$ the multiplier

Government Spending

- ▶ Caveat 1: $\chi = \frac{\alpha}{\lambda}$ no equivalence, RANK-like
- ▶ Caveat 2: above is for fixed- r . But g also has supply effects:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa (c_t + \tilde{\zeta} \tilde{g}_t) = \beta \mathbb{E}_t \pi_{t+1} + \kappa \underbrace{(c_t - c_t^*)}_{\text{output gap}}$$

Government Spending: Inequality and Tradeoffs

- γ_t *not* proportional to y gap (*unless* $\alpha = \lambda$)

$$\gamma_t = \frac{1-\chi}{1-\lambda}c_t - \left(\chi - \frac{\alpha}{\lambda}\right) \frac{\tilde{\zeta}}{1-\lambda}\tilde{g}_t = \frac{1-\chi}{1-\lambda}(c_t - c_t^*) - \frac{1-\frac{\alpha}{\lambda}}{1-\lambda}\tilde{\zeta}\tilde{g}_t.$$

- *Flex-p inequality inefficient* $\alpha \neq \lambda \rightarrow$ tradeoff (*transfer intuition*)

$$\gamma_t^* = -\frac{1-\frac{\alpha}{\lambda}}{1-\lambda}\tilde{\zeta}\tilde{g}_t$$

- Perfect redistribution/ insurance $\gamma_t^{**} = 0$:

$$c_t^{**} = c_t^* + \frac{1-\frac{\alpha}{\lambda}}{1-\chi}\tilde{\zeta}\tilde{g}_t = -\frac{\chi-\frac{\alpha}{\lambda}}{\chi-1}\tilde{\zeta}\tilde{g}_t; \quad \pi_t^{**} = \frac{\kappa}{1-\beta p} \frac{1-\frac{\alpha}{\lambda}}{1-\chi}\tilde{\zeta}\tilde{g}_t.$$

- Unless $\alpha = \lambda$, $\pi_t = 0$ inefficient (w/ perfect insurance):
conflict w/ flex-p
- Transfer transfer intuition: 3 effects, direct + 2 indirect, ...

Government Spending: Inequality and Tradeoffs

- ▶ Tension: *supply-side* π v *demand-side* γ
- ▶ Ramsey $\iff$ minimize loss (proportional to):

$$\pi_t^2 + \underbrace{\frac{\sigma^{-1} + \tilde{\varphi}}{\psi} \left(c_t + \tilde{\zeta} \tilde{g}_t \right)^2}_{\text{output gap}} + \underbrace{\frac{\sigma^{-1} \tilde{\zeta} + \tilde{\sigma}_G^{-1}}{\psi} \tilde{g}_t^2}_{\text{govt. spending}} + \underbrace{\frac{\left(\sigma \tilde{\zeta} \right)^{-1}}{\psi} \lambda (1 - \lambda) \gamma_t^2}_{\text{inequality}}$$

- ▶ g deviating from "Samuelson" $U_C^j(C) = U_G^j(G)$ Woodford 2011
- ▶ **Novel:** *inequality not* proportional to output gap wrt flex prices, unless $\alpha = \lambda$
- ▶ *Wedge:* fiscal redistribution btw welfare-relevant & flex-price output.

Optimal Policy: Exogenous Government Spending

- ▶ Optimal targeting rule, **discretion** (same as above for timeless-optimal commitment):

$$\kappa\pi_t + \frac{\sigma^{-1} + \tilde{\varphi}}{\psi} \left[1 + \frac{\lambda(\chi - 1)^2}{(1 - \lambda)\sigma\tilde{\varphi}} \right] (c_t + \tilde{\zeta}\tilde{g}_t) + \left(1 - \frac{\alpha}{\lambda} \right) \frac{(\chi - 1)\lambda\sigma^{-1}}{1 - \lambda} \frac{1}{\psi} \tilde{g}_t = 0$$

- ▶ $\alpha = \lambda$ or $\chi = 1$, optimal & feasible close y gap (wrt flex-p) and $\pi_t = 0$
 - ▶ $\rightarrow$ no inherent tradeoff, first-best: NK jargon "divine coincidence" efficient = flex-p y
- ▶ Unequal-incidence: time-varying wedge btw efficient & flex-p y

Optimal Policy: Exogenous Government Spending

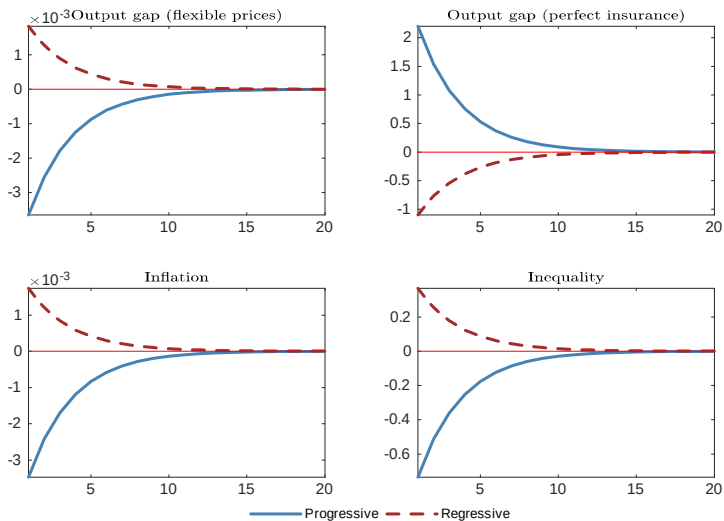
- Solution:

$$c_t + \tilde{\zeta} \tilde{g}_t = - \left(1 - \frac{\alpha}{\lambda}\right) (\chi - 1) \frac{\lambda}{1 - \lambda} \frac{\tilde{\Phi}}{1 + \sigma \tilde{\varphi}} \tilde{g}_t; \quad \pi_t = \frac{\kappa \tilde{\Phi}}{1 - \beta p} \left(c_t + \tilde{\zeta} \tilde{g}_t\right)$$

$$\gamma_t = - \left(1 - \frac{\alpha}{\lambda}\right) \left(1 + \frac{\kappa \theta}{1 - \beta p}\right) \frac{\zeta \tilde{\Phi}}{1 - \lambda} \tilde{g}_t, \quad \tilde{\Phi} > 0.$$

- $\left(1 - \frac{\alpha}{\lambda}\right) (\chi - 1)$: Key statistic for exog. & endog. redistribution
- $\alpha < \lambda \rightarrow \text{transfer} \rightarrow \gamma \downarrow$ direct + endogenously ($\chi > 1$), $c \uparrow$ further $\gamma \downarrow$
- Optimal policy: counteract, engineer deflation (NK jargon "cost-pull")
- NB: *Positive gap wrt c^{**}* (although negative wrt c^*):

$$c_t - c_t^{**} = \frac{1 - \frac{\alpha}{\lambda}}{\chi - 1} \left(1 + \frac{\kappa \theta}{1 - \beta p}\right) \tilde{\Phi} \tilde{\zeta} \tilde{g}_t.$$



Optimal-policy responses to g shock: progressive $\alpha < \lambda$ vs. regressive $\alpha > \lambda$
taxation. ($\chi > 1$)

Optimal M& F Mix: Endogenous G

- Cost-push again:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa \left(c_t + \tilde{\zeta} \tilde{g}_t \right) + u_t$$

- Optimal $\tilde{g}_t$ proportional to inequality gap:

$$\tilde{g}_t = - \left(1 - \frac{\alpha}{\lambda} \right) (\chi - 1) \frac{\lambda}{1 - \lambda} \frac{\tilde{\sigma}_G}{\tilde{\zeta} \tilde{\sigma}_G + \sigma} \left(c_t + \frac{\chi - \frac{\alpha}{\lambda}}{\chi - 1} \tilde{\zeta} \tilde{g}_t \right)$$

- Sign $\sim$ sufficient statistic $\left(1 - \frac{\alpha}{\lambda} \right) (\chi - 1)$;
 - consistent w/ f_t result, $\tilde{g}_t = 0$ w/ uniform incidence

Optimal M& F Mix: Endogenous G

- Solution:

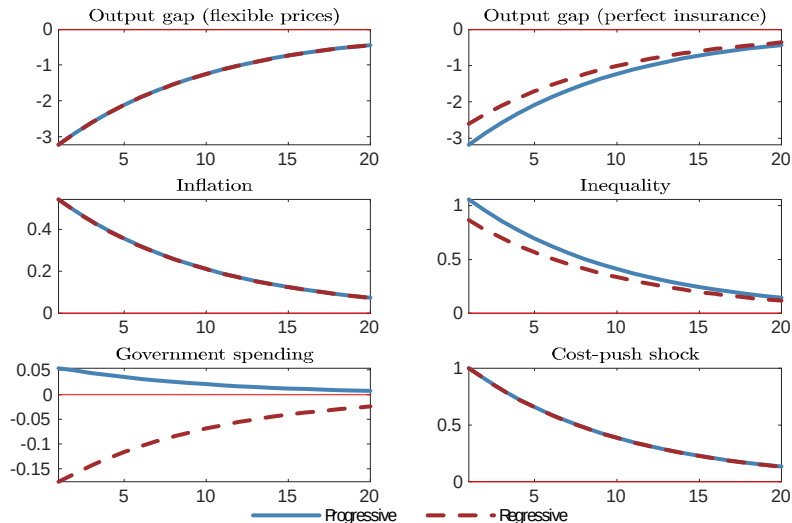
$$\tilde{g}_t^{opt} = \left(1 - \frac{\alpha}{\lambda}\right) (\chi - 1) \Psi u_t, \Psi > 0$$

$$c_t - c_t^* = c_t + \tilde{\zeta} \tilde{g}_t = - \left[\frac{1 - \lambda}{\lambda} \left(\tilde{\zeta} + \sigma \tilde{\sigma}_G^{-1} \right) + \tilde{\zeta} \left(1 - \frac{\alpha}{\lambda} \right)^2 \right] \Psi u_t$$

$$c_t - c_t^{**} = - \frac{1 - \lambda}{\lambda} \left(\tilde{\zeta} + \sigma \tilde{\sigma}_G^{-1} \right) \Psi u_t$$

$$\gamma_t = \frac{1 - \chi}{1 - \lambda} (c_t - c_t^{**}) = \frac{\chi - 1}{\lambda} \left(\tilde{\zeta} + \sigma \tilde{\sigma}_G^{-1} \right) \Psi u_t$$

- Baseline $\alpha < \lambda$ & $\chi > 1$: $c \downarrow \pi \downarrow$ optimal policy $\rightarrow$ counter-cyclical $g \uparrow$
- Accompanied by transfer to $H \rightarrow c \uparrow$ disproportionately = more effective
- g pro-cyclical with $\alpha > \lambda$ & $\chi > 1$: $g \downarrow +$ regressive $\rightarrow c \uparrow$, implicit transfer
- $\chi > 1$ empirically realistic (evidence in Patterson 2019)



Optimal-policy responses to cost-push Solid progressive ($\alpha < \lambda$), dashed regressive taxation ($\alpha > \lambda$).

Conclusions

- ▶ **Unequal incidence breaks divine coincidence**
 - ▶ monetary-fiscal policy inextricably linked in any *ANK
 - ▶ **Optimal** policy: tradeoff stabilization-redistribution
1. exog. transfers $\rightarrow$ inefficient wedges
 2. endog. transfers stab.+insurance $\rightarrow$ separation
 3. exog. spending $\rightarrow$ inefficient wedge alas unequal incidence pro/regressivity
 4. optimal spending counter- or pro-cyclical depending on redistributions

Estimated Tractable HANK-DSGE

Based on Bilbiie, Primiceri, and Tambalotti:

The Cyclicalitity of Inequality and Risk

Convergence

$$\begin{array}{c} \text{HANK} \\ \downarrow \uparrow \\ \boxed{\boxed{\text{T}_{\text{HANK}}}} \end{array}$$

Max(Micro in Macro)

s.t. Tractable

(Duality? probably not yet)

Current Work

- ▶ Estimated THANK-DSGE (w/ Primiceri and Tambalotti)
- ▶ Understanding COVID-19 (w/ Del Negro, Primiceri and Tambalotti)
- ▶ Estimated Fiscal Multipliers (w/ Crawford and Mertens)
- ▶ Fiscal Theory and Policy w/ ...
- ▶ Secular Stagnation w/ ...
- ▶ Much more yet to be done

Bilbiie (2008 JET) 4 contributions in nutshell

1. TANK analytical $\rightarrow$ income (profits') distribution key for AD&MP

Interest rate changes modify the intertemporal consumption and labor supply profile of *asset holders*, agents who smooth consumption by trading in asset markets. This affects the *real wage* and hence the demand of agents who have no asset holdings but merely consume their wage income. Variations in the real wage (marginal cost) lead to variations in profits and hence in the *dividend income of asset holders*. These variations can either reinforce (if participation is not 'too' limited) or *overturn* the initial impact of interest rates on aggregate demand. The latter case occurs if the share of non-asset holders is high enough and/or the elasticity of labor supply is low enough, for the potential variations in profit income offset the interest rate effects on the demand of asset holders. This is the main mechanism identified by this paper to change dramatically the effects of monetary policy as compared to a standard full-participation case whereby aggregate demand is completely driven by asset holders.

If participation is restricted below a certain threshold, the predictions are **strengthened**: as the share of non-asset holders increases, the link between interest rates and aggregate demand becomes stronger, and monetary policy is more effective; we label this case '*standard aggregate demand logic*', (SADL). However, when participation is restricted beyond a given threshold,

2. aggregate Euler-IS; key $c_j = \varepsilon(\lambda) * y$ (intuition Section 3.1) TANK Intro

Straightforward algebraic manipulation of the equilibrium conditions in Table 1 allows the derivation of aggregate dynamics, similar to the standard, full-participation New Keynesian benchmark. We start by deriving the *aggregate Euler equation*, or '*IS*' curve. To that end, we need to express consumption of asset holders (the only agents whose consumption obeys an Euler equation) in terms of aggregate consumption/output. Since hours of non-asset holders are constant $n_{H,t} = 0$,¹³ their consumption tracks real wage, $c_{H,t} = w_t$. Total labor supply (from the labor market clearing condition) is $n_t = [1 - \lambda] n_{S,t}$. Using these last two expressions, asset holders' labor supply equation, the production function and the goods market clearing condition into the definition of total consumption we find:

$$c_{S,t} = \delta y_t + (1 + \mu)(1 - \delta) a_t, \quad \text{where } \delta \equiv 1 - \varphi \frac{\lambda}{1 - \lambda} \frac{1}{1 + \mu}. \quad (7)$$

Substituting (7) into the Euler equation of asset holders we find the *aggregate Euler equation*, or '*IS* curve':

$$y_t = E_t y_{t+1} - \delta^{-1} [r_t - E_t \pi_{t+1}] + (1 + \mu) (1 - \delta^{-1}) [a_t - E_t a_{t+1}]. \quad (8)$$

Direct inspection of (8) suggests the impact that LAMP has on the dynamics of a standard business cycle model through modifying the elasticity of aggregate demand to real interest rates

$$w_t = \chi y_t - \varphi a_t, \quad \text{where } \chi \equiv 1 + \varphi / (1 + \mu) \geq 1 \geq \delta.$$

Bilbiie (2008 JET) 4 contributions in nutshell - cont'd

3. fiscal redistribution key for AD amplification of MP; AD elasticity:

$$c_{S,t} = \delta_\tau y_t, \quad \text{where } \delta_\tau = \frac{1}{1 - \tau^D} \left[1 - \tau^D \frac{\mu}{1 + \mu} + \frac{\tau^D - \lambda}{1 - \lambda} \frac{\varphi}{1 + \mu} \right].$$

4.3. Redistribution restores Keynesian logic

The mechanism of all the previous results relies on the interaction between labor and asset markets, namely income effects on labor supply of asset holders from the return on shares. This hints to an obvious way to restore Keynesian logic relying on a specific fiscal policy rule that shuts off this channel: tax dividend income and redistribute proceedings as transfers to non-asset holders. We focus on the IADL case whereby in the absence of fiscal policy $\delta < 0$. To make this point, consider the following simplified fiscal rule: profits are taxed at rate τ_t^D and the budget is balanced period-by-period, with total tax income $\tau_t^D D_t$ being distributed lump-sum to all non-asset holders. We focus on the case where profits are zero in steady state. The balanced-budget rule then is $\tau_t^D D_t = \lambda L_{H,t}$ which around the steady state (both profits and transfers are shares of

4. optimal policy in TANK Intro TANK

Proposition 4. *If the steady state of the model in Section 3 is **efficient** the aggregate welfare function can be approximated by (ignoring terms independent of policy and terms of order higher than 2):*

$$\begin{aligned} U_t &= -\frac{U_{CC}}{2} \frac{\varepsilon}{\psi} E_t \sum_{i=t}^{\infty} \left\{ \alpha x_{t+i}^2 + \pi_{t+i}^2 \right\}, \\ \alpha &\equiv \frac{\varphi + \gamma}{1 - \lambda} [1 - \lambda (1 - \gamma) (1 + \varphi)] \frac{\psi}{\varepsilon}. \end{aligned} \tag{18}$$